RK0

The Real-Time Kernel '0'

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Chapter 1. THE KERNEL AT A GLANCE

1.1. The design approach

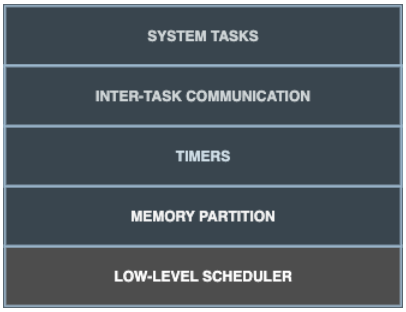
RK0 Blog: About Processes, Tasks and Threads

1.1.1. Architecture

The layered architecture can be split — roughly — into two: a top and a bottom layer. On the top, the *Executive* manages the resources needed by the application.

On the bottom, the *Low-level Scheduler* works as a software extension of the CPU.

Together, they implement the *Task* abstraction. This primitive is the *Concurrency Unit* and follows the *Thread* model. A *Task* is a *thread*.



In systems design jargon, the Executive enforces policy (what should happen). The Low-level Scheduler provides the mechanism (how it gets done). The services are the primitives that gradually translate policy decisions into concrete actions executed by the Scheduler.

RK0's goal is determinism on low-end devices. Its multitasking engine does not split user space from kernel space. Tasks execute in privileged mode and use a dedicated process stack pointer, distinct from the system stack. The rationale:

- Application tasks are not unknown entities at run time.
- Implementing system calls as traps either forces a non-preemptive kernel, or increases complexity in critical paths, degrading determinism.
- Relying on the ARMv6/7-M MPU decreases memory usage efficiency and introduces latency on control paths. It does not fit RK0's deterministic execution model.

1.1.2. Programming with RK0

RK0 is designed so that it does not get in the programmer's way.

It aims to be transparent, composable, deterministic, and with clear semantics.

Transparent means that the kernel does not hide its primitives behind opaque pointers nor introduce unexpected side effects.

Its components are composable because each feature is self-contained yet designed to work with others.

Clear semantics means mechanisms behave as expected across different usage patterns.

Determinism is pursued by avoiding execution paths whose worst case cannot be bounded or reasoned about in advance.

When possible, operations are O(1), rely on word-aligned memory, and use static allocation.

When dynamic allocation is unavoidable, its worst-case behaviour can still be characterised offline (unless the application itself prevents any meaningful offline bound).

1.1.3. Suitable Applications

Given the architecture, *RK0* targets applications with the following characteristics:

1. They are designed to handle particular devices in which real-time responsiveness is imperative.
2. Applications and middleware may be implemented alongside appropriate drivers.
3. Drivers may even include the application itself.
4. *Untested programs are not loaded*: After the software has been tested, it can be assumed reliable.

1.2. Kernel Services

RK0 has *Core Services* (always enabled) and optional services (enabled by configuration).

Core Services:

- Scheduler
- Partition Memory Allocator
- Timer Delays
- Task's Event Register

Optional Services

- Application Timer (Callouts)
- Sleep Queues
- Counting/Binary Semaphores
- Mutex Semaphores
- Message Queue (and its extension Ports)
- Most-Recent Message Protocol (MRM)

When compiled with only *Core Services*, one gets a functional Executive with less than ~3 KB ROM.

Chapter 2. Task Scheduler

RK0 employs a priority-based preemptive scheduler, aligned with a Rate-Monotonic Assignment. Tasks are typically assigned priorities according to their request rates - i.e., tasks with shorter periods are assigned to higher priorities. The highest priority is represented by the value '0'; the lowest is represented by the value '31'.

> ***RK0 Blog:*** About Real-Time, Responsiveness and Throughput

An essential characteristic of the scheduler is that it is a *preemptive run-to-completion* scheduler. This term, '*run-to-completion*' has slightly different meanings depending on the context. It is often related to strictly cooperative schedulers, in the sense tasks must *yield* the processor. Otherwise, they monopolise the CPU.

The Ready Queue table is an array of queues, which the index represents the priority

```
[][]...[] | Ready Tasks with Priority 31
[][]...[]
[][]...[]
     .
     .
     .
[][]...[] | Ready Tasks with Priority 0
```

In *RK0*, tasks with the same priority work cooperatively. This is different from schedulers that employ a *time-slice* (quantum) for round-robin: after the slice expires, the task is put at the *tail* of the *Ready Queue*.

The term *run-to-completion* here is to be interpreted as follows:

- The scheduler's behaviour is to choose the highest priority READY task to run. Always.
- The scheduler works on a First-In-First-Out discipline for tasks with the same priority.
- A task must switch to the *READY* state before being eligible for scheduling.
- A task will switch from *RUNNING* to *READY* if yielding or if being preempted by a higher priority task. Otherwise it can only go to a *WAITING* state, and eventually switch back to *READY*.
- When a task is preempted by a higher priority task, it switches from *RUNNING* to *READY* and is placed back on the *head* position of its Ready Queue. This means that it will be resumed as soon as it is the highest priority ready task again.
- On the contrary, if a task *yields*, it tells the scheduler it has completed its cycle. Then, it will be enqueued on the ready queue tail - the last queue position.
- When a task *waits* it is suspended until a condition is satisfied.
- When the condition is satisfied, it switches from *WAITING* to *READY*, and is enqueued on the tail.

- So, tasks with the same priority cooperate by either *yielding* or *waiting*.
- If a task never yields or waits, other tasks with the same or lower priority *will starve*.
- Finally, Tasks with the same priority are *initially* placed on the *Ready Queue* associated with that priority in the order they are *created*.

> *RK0* can handle context-switching with an extended frame when a float-point co-processor is available. This must be informed when compiling by defining the symbol `__FPU_PRESENT=1`.

2.1. Scheduler Design Internals

A notable scheduler characteristic is constant-time complexity (*O(1)*) with low latency. This was achieved by carefully composing the data structures and an efficient *'choose-next'* algorithm, detailed below.

2.2. Scheduler Data Structures

Task Control Block

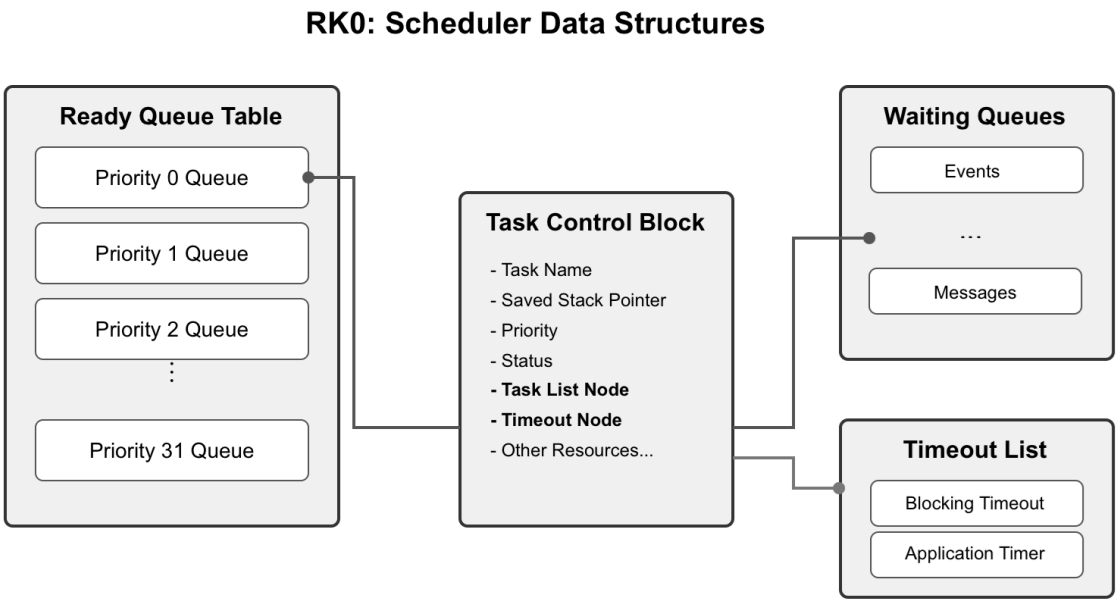
Every primitive is associated to a data structure we refer to as its *Control Block*. A Task Control Block is a record for stack, resources, and time management. The table below partially represents a Task Control Block (as this document is live, this might not reflect the exact fields of the current version).

Task Control Block
Task name
Task ID
Status
Assigned Priority
Effective Priority
Saved Stack Pointer
Stack Address
Stack Size
Event Register Control Block
Last wake-time
Next wake-time
Time-out Flag
Preemption Flag
Owned Resources List
Waiting Resources List
Timeout List Node

Task Control Block
TCB List Node

Tasks are static — they are not created (or destroyed) on runtime. There is no fork or join.

In practice, tasks are either *RUNNING* or '*waiting*' for their turn to run.



We need to define *WAITING* and *READY* clearly:

- A *READY* task will be dispatched; therefore, switch to *RUNNING* whenever it is the highest priority *READY* task.
- A WAITING task depends on a condition—generalised as an event—to transition to the READY state.
- Events that are not strictly time-related can be bounded by an upper limit of N ticks, or can be unbounded (the task waits indefinitely, i.e. “forever”). A task may also check for an event without blocking at all, using `try` semantics.
- Logically, the *WAITING* state will assume different pseudo-states related to the kind of event that will switch a task to *READY*:
- SLEEPING: the task is either waiting for an event associated with an `RK_SLEEP_QUEUE` object, or sleeping for a given amount of time with no explicit kernel object associated. The various `SLEEPING_*` states are explained throughout this document.
- PENDING: the task has suspended itself, waiting for a combination of Event Flags on its Event Register. When this combination is satisfied, it switches to READY.
- BLOCKED: the task is blocked on a mutex or semaphore. When a token becomes available, or the mutex is unlocked, it switches to READY.
- SENDING/RECEIVING: a producer task, when blocking on a Message Passing object, switches its status to SENDING; a consumer switches to RECEIVING. When, respectively, a slot or a message becomes available, the task switches to READY.

The scheduler rules, not the heap.

RK0 tasks are static.

It's a design decision rooted in real-time correctness.

Besides an application-specific system software does not need to treat tasks as 'unknown' objects.

The wins:

- A memory layout the systems programmer knows.
- No alignment traps.
- Link-time visibility:
  - Each task's stack is a named symbol in the linker map.
  - You can inspect and verify the memory layout before flashing.
  - A simple `objdump` reveals all stack allocations — that's peace of mind.

Task Queues

The backbone of the queues where tasks will wait for their turn to run is a circular doubly linked list: removing any item from a double list takes O(1) (provided we don't need to search the item). As the kernel knows each task's address, adding and removing is always O(1). Singly linked lists can't achieve O(1) for removal.

Ready Queue Table

Another design choice to achieve O(1) is the global ready queue, which is a table of FIFO queues—each queue dedicated to a priority—and not a single ordered queue. So, enqueuing a ready task is always O(1). Given the sorting needed, the time complexity would be O(n) if tasks were placed on a single ready queue.

Waiting Queues

The scheduler does not have a unique waiting queue. Every kernel object that can block a task has an associated waiting queue. Because these queues are a scheduler component, *they follow a priority discipline*: the highest priority task is dequeued first, *always*.

When an event capable of switching tasks from *WAITING* to *READY* happens, one or more tasks (depending on the mechanism) are then placed on the ready list, unique to their priority. Now, they are waiting to be picked by the scheduler—that is the definition of *READY*.

The scheduling algorithm

As the ready queue table is indexed by priority - the index 0 points to the queue of ready tasks with priority 0, and so forth, and there are 32 possible priorities - a 32-bit integer can represent the state of the ready queue table. It is a BITMAP:

```
The BITMAP computation: ((1a) OR (1b)) AND (2), s.t.:

(1a) Every Time a task is readied, update: BITMAP |= (1U << task->priority );
(1b) Every Time an empty READY QUEUE becomes non-empty, update: BITMAP |= (1U <<
queueIndex)
(2): Every Time READY QUEUE becomes empty, update: BITMAP &= ~(1U << queueIndex);
EXAMPLE:

  Ready Queue Index :     (6)5 4 3 2 1 0
          Not empty :      1 1 1 0 0 1 0
                           ------------->
                 (LOW)  Effective Priority  (HIGH)
In this case, the scenario is a system with 7 priority task levels. Queues with
priorities 6, 5, 4, and 1 are not empty.
```

Having the Ready Queue Table bitmap, we find the highest priority non-empty task list as follows:

(1) Isolate the **rightmost** '1':

```
RBITMAP = BITMAP & -BITMAP. (- is the bitwise operator for two's complement: ~BITMAP +
1) `
```

In this case:

```
                          [31]       [0]  :  Bit Position
                            0...1110010   :  BITMAP
                            1...0001110   : -BITMAP
                           =============
                            0...0000010   :  RBITMAP
                                    [1]
```

The rationale here is that, for a number N, its 2's complement -N, flips all bits - except the rightmost '1' (by adding '1') . Then, N & -N results in a word with all 0-bits except for the less significant '1'.

(2) Extract the **rightmost '1'** ***position***:

- For ARMv7M, we benefit from the `CLZ` instruction to count the *leading zeroes*. As they are the number of zeroes on the left of the rightmost bit, '1', this value is subtracted from 31 to find the Ready Queue index.

```
RK_FORCE_INLINE static inline
unsigned __getReadyPrio(unsigned readyQBitmap)
{
    unsigned ret;
    __ASM volatile (
        "clz    %0, %1     \n"
        "neg    %0, %0     \n"
```

```
        "add    %0, %0, #31\n"
        : "=&r" (ret)
        : "r" (readyQBitmap)
        :
    );
    return (ret);
}
```

This instruction would return #30, and #31 - #30 = #01 in the example above.

- For ARMv6M there is no suitable hardware instruction. The algorithm is written in C and counts the *trailing zeroes*, thus, the index number. Although it might vary depending on your compiler settings, it takes ~11 cycles (*note it is still O(1)*):

```
/*
  De Brujin's multiply+LUT
  (Hacker's Delight book)
*/

/* table is on a ram section  for efficiency */
RK_SECTION(getReadyTable)
const static unsigned readyPrioTbl[32] =
{
 0, 1, 28, 2, 29, 14, 24, 3, 30, 22, 20, 15, 25, 17, 4, 8,
 31, 27, 13, 23, 21, 19, 16, 7, 26, 12, 18, 6, 11, 5, 10, 9
};

RK_FORCE_INLINE static inline
unsigned __getReadyPrio(unsigned readyQBitmap)
{
    unsigned mult = readyQBitmap * 0x077CB531U;

    /* Shift right the top 5 bits
     */
    unsigned idx = (mult >> 27);

    /* LUT */
    unsigned ret = (unsigned)readyPrioTbl[idx];
    return (ret);
}
```

For the example above, `mult = 0x2 * 0x077CB531 = 0x0EF96A62`. The 5 leftmost bits (the index) are `00001` → `table[1] = 1`.

During a context switch, the procedures to find the highest priority non-empty ready queue table index are as follows:

```
static inline RK_PRIO kCalcNextTaskPrio_(VOID)
{
```

```
    if (readyQBitMask == 0U)
    {
        return (idleTaskPrio);
    }
    readyQRightMask = readyQBitMask & -readyQBitMask;
    RK_PRIO prioVal = (RK_PRIO) (__getReadyPrio(readyQRightMask));
    return (prioVal);
}

VOID kSwtch(VOID)
{
    /* O(1) complexity */
    nextTaskPrio = kCalcNextTaskPrio_();

    RK_TCB* nextRunPtr = NULL;

    /* O(1) complexity */
    kTCBQDeq(&readyQueue[nextTaskPrio], &nextRunPtr);

    runPtr = nextRunPtr;

}
```

Chapter 3. Timers and Delays

3.1. Busy delay

A busy-wait delay `kBusyDelay(t)` keeps a task spinning for `t` ticks. That is, the task does nothing but does not suspend or yield (but can be preempted). This service finds its use when simulating workloads.

> Context switching is probably the most significant overhead on a kernel. The time spent on the System Tick handler contributes to much of this overhead.
>
> Design Choice:
>
> - Timers are kept on a single list; only the head element needs to be updated using a delta-queue approach.
> - Application Timers that trigger callbacks are run on a deferred, non-preemptible system task.
>
> Benefits:
>
> - Keep the overhead of updating timers as minimal as possible with the delta queue;
> - Deferring the Application Timer to a high-priority, non-preemptible system task meet the requested callback period while keeping the ability to track system ticks.

Timeout Node
Timeout Type
Absolute Interval (Ticks)
Relative Interval (Ticks)
Waiting Queue Address
Next Timeout Node
Previous Timeout Node

Every task is prone to events triggered by timers described in this section. Every Task Control Block has a node to *a timeout list*. This list is doubly linked treated a delta-sequence.

A set T_{set} `= {(T1,8), (T2,6), (T3,10)}` will be started at a relative time `0` as a sequence T_{seq} `= <(T2,6), (T1,2), (T3,2)>`.

Thus, for every system tick, only the head element on the list needs to be decreased — yielding O(1) on *decreasing*, that happens on the Hardware interrupt for the *System Tick*.

The ordering for the delta-queue is not O(1), it is O(n). A decrease happens on every SysTick interrupt; and the ordering happens only when adding a new node to the list.

3.2. Sleep Timers

There are three `sleep` primitives in RK0 — they behave differently.

3.2.1. Sleep Delay

The `sleepdelay()` (aliased as `sleep()`) puts a task to sleep for the exact number of `t` ticks on every call — no matter when the last call has happened.

Example:

```
VOID Task1(VOID* args)
{
    RK_UNUSEARGS
    UINT count = 0;
    while (1)
    {

        logPost("Task1: sleep");
        kSleep(300);
        /* wake here */
        count += 1U;
        if (count >= 5)
        {
            kDelay(25); /* spin */
            count=0;
            /* every 5 activations there will be a drift */
        }
    }
}
```

Output:

```
0 ms :: Task1: sleep
300 ms :: Task1: sleep  <-- +300
600 ms :: Task1: sleep  <-- +300
900 ms :: Task1: sleep  <-- +300
1200 ms :: Task1: sleep <-- +300
1525 ms :: Task1: sleep <-- +325
1825 ms :: Task1: sleep <-- +300
2125 ms :: Task1: sleep <-- +300
2425 ms :: Task1: sleep
2725 ms :: Task1: sleep
3050 ms :: Task1: sleep
3350 ms :: Task1: sleep
3650 ms :: Task1: sleep
3950 ms :: Task1: sleep
4250 ms :: Task1: sleep
```

```
4575 ms :: Task1: sleep
```

3.2.2. Compensated Sleep Delays

These are suspensions that recompute the time considering the drift between calls. They are typically used to create periodic tasks with explicit periods. The general pattern is:

```
VOID Task(VOID *args)
{
    <initialisation>

    while(1)
    {
        <periodic code>; /* this has an execution time */
        scheduleNext(PERIOD); /* cycle is finished: compute delay to keep PERIOD */
    }
}
```

The RMS algorithm considers all tasks have a common phase grid, that is when they are made READY, or are eligible to be scheduled. This means that from the very first activation, the time elapsed is already take into account to when the next release should happen.

Nevertheless, it is common for kernels to provide sleep primitives that take into account a local anchor time normally set on the `<initialisation>` code — and will work as `waitUntil(&anchor, PERIOD)`. RK0 provides both, and they suit different cases, as exposed below.

3.2.2.1. Periodic Release Sleep

`sleeprelease(P)` is used to delay a task so it is released on periodic rate. The sleep time is recalculated by the kernel on every call.

If the task wakes late by `N` ticks with $0 < N < P$, the kernel compensates by scheduling the next wake earlier (shortening the next sleep) so that over two periods the phase is preserved:

Say a task is expected to return from its k_{eth} sleep at $T_{k+1} = T_k + P$ [ticks]. If the task is resumed at $T_{k+1} = T_k + P + N$, upon detecting this drift, the kernel sets: $(T_{k+2} = T_{k+1} + P - N)$ for $N < P$.

This can be rewritten as:

$(T_{k+2} = T_k + P + N + P - N) \leftrightarrow (T_{k+2} - T_k = 2P)$

Example:

```
VOID Task1(VOID* args)
{
    RK_UNUSEARGS
    UINT count = 0;
    while (1)
    {
```

```
        logPost("Task1 released.");

         count += 1U;
         if (count >= 5)
         {
             kDelay(25); /* spin */
             count=0;
         }

         kSleepRelease(300); /*P=300 ticks; tick=1ms*/


    }
}
```

Output:

```
.
.

/* R is release time */

1200 ms :: Task1: sleep periodic (R==4P)        (n)
1525 ms :: Task1: sleep periodic (5P<R<6P)       |
1800 ms :: Task1: sleep periodic (6P)            |
2100 ms :: Task1: sleep periodic (7P)            |
2400 ms :: Task1: sleep periodic (8P)            |
2700 ms :: Task1: sleep periodic (9P)            |
3025 ms :: Task1: sleep periodic (10P<R<11P)     |
3300 ms :: Task1: sleep periodic (11P)           |
3600 ms :: Task1: sleep periodic (12P)           |
3900 ms :: Task1: sleep periodic (13P)           |
4200 ms :: Task1: sleep periodic (14P)           |
4525 ms :: Task1: sleep periodic (15P<R<16P)     |
4800 ms :: Task1: sleep periodic (16P)          (m)  m-n=12
.                                              -----
.                                           Phase=3600=12xP
.
```

This mechanism is phase-locked. When the lateness is greater or equal to `P` it *skips* one or more releases to stay locked to the phase grid. In some sense, the `period` value can be seen as a *deadline* — if not met, the scheduler rejects to run on that activation.

`sleeprelease()` makes easier to perform worst-response time analysis on periodic tasks.

> A set of periodic tasks must have priorities assigned properly (highest request rate, highest priority — the lower the period, the higher the priority). For `sleeprelease()` this is mandatory given the common phase grid.

3.2.2.2. Sleep Until (local anchor base)

`sleepuntil(anchor, period)` is somehow similar to `sleeprelease()`, but differs in two important aspects:

- The reference used to calculate how long to suspend to keep its rate is local to each task. It means the time before the first time the task is dispatched is dismissed.
- A late release longer than 1 period will return and run immediately. It prioritises execution count within a time-window - not the phase across releases.

The snippet belows clearly demonstrates how each mechanism handles lateness that are longer than 1 period:

```
/* Every 3rd call both tasks will add up a delay longer than the task's period */

VOID HTask(VOID* args) /* higher priority: Period is 300 ticks */
{
    RK_UNUSEARGS
    UINT count = 0;
    while (1)
    {

        logPost("Higher: begin\r\n");
        /* wake here */
        count += 1U;
        kDelay(5);
        if (count >= 3)
        {
            kSleep(400); /* suspend */
            count=0;
        }
        logPost("Higher: end\r\n");
        kSleepRelease(300);

    }
}


VOID LTask(VOID *args) /* lower priority: Period is 400 ticks */
{
    RK_UNUSEARGS
    RK_TICK anchor = kTickGet();
    UINT count=0;
    while (1)
    {

        logPost("Lower: begin\r\n");
        /* wake here */
        count += 1U;
```

```
        kDelay(5);
        if (count >= 3)
        {
            kSleep(500);
            count=0;
        }
        logPost("Lower: end\r\n");
        kSleepUntil(&anchor, 400);
    }
}
```

Output:

```
       0 ms :: Higher: begin
       5 ms :: Higher: end
       5 ms :: Lower: begin
      10 ms :: Lower: end
     300 ms :: Higher: begin
     305 ms :: Higher: end
     405 ms :: Lower: begin
     410 ms :: Lower: end

    /* H 3rd run, expected next at 900ms */
     600 ms :: Higher: begin


   /* L 3rd run, expected next at 1205 ms */
     805 ms :: Lower: begin

    /* H Drift: 1005ms - 600ms = 405 ms > 300ms */
    1005 ms :: Higher end

    /* H released again @ next multiple of 300. */
    1200 ms :: Higher: begin
    1205 ms :: Higher: end

   /* L Drift: 1310ms - 805ms = 505 ms > 400ms */
    1310 ms :: Lower: end
    /* it runs again  immediately */
    1310 ms :: Lower: begin
```

One normally does not write a code with periodic tasks expecting they will not keep their rate. But on the field a transient overload might cause it to happen. If it does, you choose the policy that is best-fit for your task: preserve phase (skip) or preserve execution count.

3.3. Blocking Time-out

These are internal timers associated with kernel calls that are blocking. Thus, establishing an upper-bound waiting time might benefit them. When the time for unblocking is up, the kernel call returns, indicating a timeout. This value is passed as a number of ticks.

When blocking is associated with a kernel object (other than the Task Control Block), the timeout node will store the object waiting for queue's address, so it can be removed if time expires.

A kernel call is made non-blocking, that is *try semantics*, by assigning the value `RK_NO_WAIT`, the function returns immediately if unsuccessful. The value `RK_WAIT_FOREVER` suspends a task indefinitely until the condition is satisfied. Timeout values above `RK_MAX_PERIOD` are invalid.

In practice, we often block either using `RK_WAIT_FOREVER` or do not block (*try semantics*, `RK_NO_WAIT`).

Use a bounded timeout only when you expect occasional misses and you know how to handle them.

> Importantly, *an ISR shall **never** block*. Any blocking call from an ISR is invalid and triggers fault handling when error checking is enabled.

3.4. Callout Timers (Application Timers)

Timer Control Block
Option: Reload/One-Shot
Phase (Initial Delay)
Callout Function Pointer
Callout Argument
Timeout Node

These are Application Timers that will issue a callback when expiring. In addition to a callout function, an Application Timer receives an initial phase delay and a period and can choose to run once (one-shot) or auto-reload itself.

The callback runs within a System Task with priority 0 and is non-preemptible, which makes the scheduler prioritise it over other tasks. Callouts must be short and unblocking, as they can cause high CPU contention.

Application Timers (with autoreload) will keep track of delays in between activations.

3.5. System Tick

A dedicated peripheral that generates an interrupt after a defined period provides the kernel time reference. For ARMv6/7M, this peripheral is the built-in SysTick, a 24-bit counter timer. The handler performs some housekeeping on every tick and assesses the need to call a context switch.

The 'housekeeping' accounts for global timer tracking and any tick-dependent condition that might

change a task status. When a timer expires, it may switch a task from `WAITING` to `READY` or schedule a callback. For callbacks, execution is deferred to the `PostProcSysTask` system task, where callbacks run and timer state is updated.

Note that tasks might switch from `WAITING` to `READY` for reasons other than tick-related. In these cases, context switching might be triggered immediately if the readied task can preempt the running task.

3.6. System Tasks

System Tasks are tasks that perform housekeeping/system related procedures outside interrupt handlers.

Currently there are two the *Idle Task* and the *PostProcSysTask*.

The *PostProcSysTask* is used to execute *Application Timer* callbacks and deferred some other deferred procedures such as the broadcast signals on Sleep Queues or Semaphores when issued from ISRs. The idea is the same, keep ISR shorts. It does not mean that this task can be expensive though because, *it cannot be preempted* by user tasks, so timer callouts must be short.

> Keep application timer callbackks minimal and non blocking. Avoid flushing from ISR.

The *Idle Task* runs whenever there is no other ready task to be dispatched. The CPU enters on *low-power*. If `RK_CONF_MIN_PRIO` is set to 31 — and it can even if not all priorities are occupied (noting there will be a small memory overhead) — the IdleTask is dispatched when the Ready bitmap is 0x00000000.

Chapter 4. Memory Allocator

Memory Allocator Control Block
Associated Block Pool
Number of Blocks
Block Size
Number of Free Blocks
Free Block List

The standard C library `malloc()` leads to fragmentation and (also, because of that) is highly indeterministic. Unless we use it once - to allocate memory before starting up, it doesn't fit. But often, we need to 'multiplex' memory amongst tasks over time, that is, to dynamically allocate and deallocate.

To avoid fragmentation, we use fixed-size memory blocks. A simple approach would be a static table marking each block as free or taken. With this pattern, you will need to 'search' for the next available block, if any - the time for searching changes - bounding this search to a maximum number of blocks, or *O(n)*. To optimise, an approach is to keep track of what is free using a dynamic table—a linked list of addresses. Now we have *O(1)*.

We use "meta-data" to initialise the linked list. Every address holds the "next" address value. All addresses are within the range of a pool of fixed-size blocks. This approach limits the minimal size of a block to the size of a memory address—32 bits for our supported architecture.

Yet, this is the cheapest way to store meta-data. If not stored on the empty address itself, an extra 32-bit variable would be needed for each block, so it could have a size of less than 32 bits.

> Allocating memory at runtime is a major source of latency (1), indeterministic (2) behaviour, and footprint overhead (3).
>
> Design choice: the allocator's design achieves low-cost, deterministic, fragmentation-free memory management by using fixed-size word-aligned block sizes (1)(2) and embedding metadata within the memory blocks themselves (3).
>
> Benefits: Run-time memory allocation benefits have no real-time drawbacks.

> The kernel will always round up the block size to the next multiple of 4. Say the user creates a memory pool, assigning blocks to be 6-byte wide; they will turn into 8-byte blocks.

4.1. How it works

When a routine calls `alloc()`, the address to be returned is the one a "free list" is pointing to, say `addr1`. Before returning `addr1` to the caller, we update the free list to point to the value stored within `addr1` - say `addr8` at that moment.

When a routine calls `free(addr1)`, we overwrite whatever has been written in addr1 with the value-free list point to (if no more `alloc()` were issued, it would still be `addr8`), and `addr1` becomes the free list head again.

Allocating and deallocating fixed-size blocks using this structure and storing meta-data this way is as deterministic (*O(1)*) and economical as we can get for dynamic memory allocation.

Chapter 5. *Inter-Task Communication*

RK0 Blog:

- About Inter-Task Communication - Part 1
- About Inter-Task Communication - Part 2

Inter-Task Communication (ITC) refers to the mechanisms that enable tasks to coordinate/cooperate/synchronise by means of sending or receiving information that falls into two logical categories: *Signals* or *Messages*.

- ***Signals***: A *Signal is either present or absent*. Its meaning is implicit.
- ***Messages***: A *Message* is a means of coordinating and exchanging information altogether. Different from a *Signal*, a Message needs to be parsed (interpreted).

5.1. Task Events (Task Event Flags)

Within Task Control Block
Event Register (32 flags)
Required Signal Flags
Options (ALL/ANY)

Each Task Control Block stores a 32-bit event register (one event per bit). A set bit means an event is pending.

The API is bit-mask oriented:

- `kTaskEventSet(task, mask)` posts flags with OR semantics.
- `kTaskEventGet(required, RK_EVENT_ANY|RK_EVENT_ALL, &got, timeout)` waits until the required pattern is met.
- `kTaskEventQuery(task, &flags)` reads current flags.
- `kTaskEventClear(task, mask)` clears selected flags.

If the required pattern is not met, the task can block (logical state: `RK_PENDING`) until timeout or until another task posts matching flags.

On successful `get()`, the matched required positions are cleared from the task event register.

`0x00` is invalid as a `set()` mask and as `required` for `get()`.

5.1.1. Usage Example: Supervisor Task

One common pattern is to start each loop by checking pending events. In a supervisor task, this becomes an event hub.

```
/* events */
#define CO2_HIGH_EVENT          RK_EVENT_01
#define O2_LOW_EVENT            RK_EVENT_02
#define HUM_HIGH_EVENT          RK_EVENT_03
#define TEMP_HIG_EVENT          RK_EVENT_04



/* task handles */
#define TEMPERATURE_TASK        tempCtl
#define PRESSURE_TASK           pressCt
#define HUM_TASK                humCtl


VOID SupervisorTask(VOID *args)
{
    RK_UNUSEARGS;

    while(1)

    {
        ULONG gotFlags = 0UL;

        RK_ERR err = kTaskEventGet(0xFFFF,
                                   RK_FLAGS_ANY,
                                   &gotFlags,
                                   SUPERVISOR_T_PERIOD);

        if (err == RK_ERR_SUCCESS && gotFlags != 0)
        {
                switch (gotFlags)
                {

                  /* map  events combination to signal handlers */



                }
            }
        }

        /* if there is anything to do if time out */
    }
}
```

Task Events are the only ITC primitive that cannot be disabled; they are a *Core Mechanism*.

> RK0 does not implement BSD/UNIX-like asynchronous signals (POSIX style).
>
> Tasks explicitly wait (`get()`) for bit patterns, and others `set()` bit patterns, making

> the scheduler aware of all blocking conditions.
>
> Thus, unlike traditional asynchronous signal mechanisms (as BSD/UNIX Signals), no code is ever executed at an arbitrary instruction boundary.

5.2. Semaphore

Semaphore Control Block
Counter (Unsigned Integer)
Maximum Value
Waiting Queue

> A semaphore S is a nonnegative integer variable, apart from the operations it is subjected to. S is initialized to a nonnegative value. The two operations, called P and V, are defined as follows:
>
> `P(S): if S > 0 then S := S-1, else the process is suspended until S > 0.`
>
> `V(S): if there are processes waiting, then one of them is resumed; else S := S+1.`
>
> (Dijkstra, 1968)

Semaphores are *public* kernel objects for signalling and waiting on countable events.

`V()` in RK0 semaphores maps to `post()` and `P()` to `pend()`.

5.2.1. Counting Semaphore and Binary Semaphores

The typical use case for semaphores is as a "credit tracker": use `pend()` to consume a credit and `post()` to return a credit (for example, free slots in a queue). These are *Counting Semaphores*.

A *Binary Semaphore* is a counting semaphore with maximum value `1`: the state is either available or unavailable.

Binary semaphores are often used for task-to-task or ISR-to-task synchronisation, and sometimes for mutual exclusion (with caveats discussed later).

5.2.2. Semaphores in RK0

To initialise a semaphore in RK0, provide two values: initial count and maximum count. When the counter is at maximum, `post()` does not increment it and returns `RK_ERR_SEMA_FULL`.

A *Binary Semaphore* is therefore created by setting maximum count to `1`. A counting semaphore that is intended never to saturate can use `UINT32_MAX`.

Besides `init()`, `post()`, and `pend()`, `query()` inspects current state: non-negative means current count; negative means number of tasks waiting.

5.2.3. Flushing or Broadcast

A `flush()` is a *broadcast* signal: it releases all tasks currently pending on the semaphore. If successful, the semaphore counter remains `0`.

> Flushing/broadcasting must be used with care. Releasing many tasks at once can increase scheduling pressure and hurt response time.

Flushing is allowed from ISRs but you should have a good reason. Nevertheless, they are deferred to the *PostProcSysTask* - so they will happen asynchronously.

Bounded Buffer with Semaphores

Items are buffered within a memory region whose capacity is `K` items.

Thus: `0 ≤ (Number of Inserted) − (Number of Extracted) ≤ K`.

Using semaphores the pattern is as follows:

1. A semaphore with `K` tokens to track free slots, preventing producers from proceeding when there are no free slots.
2. Another semaphore, with `K` tokens, for the number of items, not allowing consumers to proceed if there are no items.
3. A 1-token semaphore so only one task manipulates the buffer at a time.

```
/* a ring buffer of items */
#define BUFSIZ (K)
static ITEM_t buf[BUFSIZ]={0};
static UINT getIdx = 0U;
static UINT putIdx = 0U;
/* this indexes==0 could either mean FULL or EMPTY for a regular
circular buffer with wrap-around.
With semaphores the state is well defined.
*/

RK_SEMAPHORE  itemSema;
RK_SEMAPHORE  slotSema;
RK_SEMAPHORE  acquireSema;


VOID kApplicationInit(VOID)
{

  /*buffer is initialised empty */
    kSemaphoreInit
    (   &itemSema,
        0,   /* no item  */
        K    /*max items */
    );
```

```
    kSemaphoreInit
    (   &slotSema,
        K, /* K free slots */
        K  /* max slots */
    );

    /* and free */
    kSemaphoreInit
    (   &acquireSema,
        1, /* free to access */
        1  /* 1 max task allowed */
    );


VOID PutItem(ITEM_t* insertItemPtr)
{
    /* wait for room */
    kSemaphorePend(&slotSema, RK_WAIT_FOREVER);

    /* wait for availability */

    kSemaphorePend(&acquireSema,  RK_WAIT_FOREVER);
    buf[putIdx] = *insertItemPtr;
    putIdx += 1U; putIdx %= BUFSIZ;
    /* signal availability */
    kSemaphorePost(&acquireSema);

    /* signal item */
    kSemaphorePost(&itemSema);
}


 VOID GetItem(ITEM_t* extractItemPtr )
{

    /* wait for an item */
    kSemaphorePend(&item, RK_WAIT_FOREVER);

    /* wait for availability */
    kSemaphorePend(&acquireSema,  RK_WAIT_FOREVER);

    *extractItemPtr = buf[getIdx];
    getIdx+=1U; getIdx %= BUFSIZ;

    /* signal availability */
    kSemaphorePost(&acquireSema);

    /* signal room */
    kSemaphorePost(&slotSema);
```

```
}
```

The solution above has Put() and Get() as blocking methods.

If the producer and the consumer run at different rates, eventually, they will synchronise to the lowest rate.

The numbers below are from a run with a buffer of 32 items (integers being incremented are the produced data).

The producer is twice faster than the consumer. Initially at every 2 insertions there is a single remove. Eventually, the buffer is filled up, and tasks run at lockstep (put, get, put, get...), at the consumer pace:

```
Put 59 <-
Put 60 <-
------
Got 30 ->
------
Put 61 <-
Put 62 <-
------
Got 31 ->
------
Put 63 <-
Put 64 <-
--------
Got 32  | ->
Put 65  . <-
       <x>[Full Queue, Producer blocks]
Got 33  | ->  [Consumer unblocks producer...]
Put 66  . <-
       <x>[Full Queue]
Got 34  | ->  [Consumer unblocks producer...]
Put 67  . <-
       <x>[Full Queue]
```

5.2.4. Mutex Semaphore (Locks)

Mutex Control Block
Locked State (Boolean)
Owner
Protocol Flag (`RK_NO_INHERIT` / `RK_INHERIT`)
Waiting Queue

Mutex Control Block
Mutex Node (list node within the owner TCB)

Some regions are critical and must not be executed by more than one task at once. Acquiring (`lock()`) a mutex before entering and releasing (`unlock()`) after leaving makes the region mutually exclusive.

A *Mutex* is a binary semaphore with ownership: once a task locks a mutex only that task can unlock it.

If a task tries to acquire a locked mutex, it switches to `BLOCKED` until the owner unlocks it. When released, the highest-priority waiter is dequeued first. Unlike semaphores, unlocking by a non-owner is invalid and rejected.

Mutexes are only for mutual exclusion; they are not signalling primitives.

PS: RK0 mutexes are non-recursive. Re-entrant locking of the same mutex returns `RK_ERR_MUTEX_REC_LOCK` *and is considered a fault.*

5.2.4.1. Priority Inversion and PIP

Let TH, TM, and TL be three tasks with priority high (H), medium (M) and low (L), respectively. Say TH is dispatched and blocks on a mutex that 'TL' has acquired (i.e.: *"TL is blocking TH"*).

If 'TM' does not need the resource, it will run and preempt 'TL'. And, by transition, 'TH'.

From now on, 'TH' has an *unbounded waiting time* because any task with priority higher than 'L' that does not need the resource indirectly prevents it from being unblocked — *awful.*

The Priority Inheritance (PI) Protocol avoids this unbounded waiting. It is characterised by an invariant, simply put:

> PIP Invariant: *At any instant a Task assumes the highest priority among the tasks it is blocking.*

If employed on the situation described above, task TM cannot preempt TL, whose effective priority would have been raised to 'H'.

It is straightforward to reason about this when you consider the scenario of a single mutex.

When locks nest, the protocol also needs to be:

- Transitive: if T1 blocks T2 and T2 blocks T3, the highest priority (T3) must propagate back to T1 through T2.

This is the hard part of a correct implementation: updates must preserve the invariant across changing wait chains and multiple mutexes.

This blog shows an even more intricate case of priority inversion handling.

RK0 Blog: Transitive Priority Inheritance on RK0 mutexes

Below, a PIP-only case in which locks nest:

```
/* Task1 has the Highest nominal priority */
/* Task2 has the Medium nominal priority */
/* Task3 has Lowest nominal priority */

/* Note Task3 starts as 1 and 2 are delayed */

RK_DECLARE_TASK(task1Handle, Task1, stack1, STACKSIZE)
RK_DECLARE_TASK(task2Handle, Task2, stack2, STACKSIZE)
RK_DECLARE_TASK(task3Handle, Task3, stack3, STACKSIZE)


RK_MUTEX mutexA;
RK_MUTEX mutexB;

VOID kApplicationInit(VOID)
{
    K_ASSERT(!kCreateTask(&task1Handle, Task1, RK_NO_ARGS, "Task1", stack1, \
        STACKSIZE, 1, RK_PREEMPT));
    K_ASSERT(!kCreateTask(&task2Handle, Task2, RK_NO_ARGS, "Task2", stack2, \
        STACKSIZE, 2, RK_PREEMPT));
    K_ASSERT(!kCreateTask(&task3Handle, Task3, RK_NO_ARGS, "Task3", stack3, \
        STACKSIZE, 3, RK_PREEMPT));

/* mutexes initialised with priority inheritance enabled */
    kMutexInit(&mutexA, RK_INHERIT);
    kMutexInit(&mutexB, RK_INHERIT);
}



VOID Task3(VOID *args)
{
    RK_UNUSEARGS
    while (1)
    {
        printf("@ %lums: [TL] Attempting to LOCK 'A' | Eff: %d | Nom: %d\r\n",
kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kMutexLock(&mutexA, RK_WAIT_FOREVER);

        printf("@ %lums: [TL] LOCKED 'A' (in CS) | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);
```

```
        kDelay(60); /* <-- important */

        printf("@%lums: [TL] About to UNLOCK 'A' | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kMutexUnlock(&mutexA);

        printf("--->");
        printf("@%lums: [TL] Exit CS | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kSleep(4);
    }
}

VOID Task2(VOID *args)
{
    RK_UNUSEARGS
    while (1)
    {
        kSleep(5);

        printf("@%lums: [TM] Attempting to LOCK 'B' | Eff: %d | Nom: %d\r\n",
kTickGet(),
               runPtr->priority, runPtr->prioNominal);
        kMutexLock(&mutexB, RK_WAIT_FOREVER);

        printf("@%lums: [TM] LOCKED 'B', now trying to LOCK 'A' | Eff: %d | Nom:
%d\r\n",
               kTickGet(), runPtr->priority, runPtr->prioNominal);
        kMutexLock(&mutexA, RK_WAIT_FOREVER);

        printf("@%lums: [TM] LOCKED 'A' (in CS) | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);
        kMutexUnlock(&mutexA);

        printf("@%lums: [TM] UNLOCKING 'B' | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kMutexUnlock(&mutexB);

        printf("--->");

        printf("@%lums: [TM] Exit CS | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);
    }
}

VOID Task1(VOID *args)
{
    RK_UNUSEARGS
```

```
    while (1)
    {
        kSleep(2);

        printf("@%lums: [TH] Attempting to LOCK 'B'| Eff: %d | Nom: %d\r\n",
kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kMutexLock(&mutexB, RK_WAIT_FOREVER);

        printf("@%lums: [TH] LOCKED 'B' (in CS)  | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);

        kMutexUnlock(&mutexB);

        printf("--->");

        printf("@%lums: [TH] Exit CS | Eff: %d | Nom: %d\r\n", kTickGet(),
               runPtr->priority, runPtr->prioNominal);
    }
}
```

Result and comments:

```
>>>> TL locks 'A'. Higher priority tasks are sleeping. <<<<

@ 14720ms: [TL] Attempting to LOCK 'A' | Eff: 3 | Nom: 3
@ 14720ms: [TL] LOCKED 'A' (in CS) | Eff: 3 | Nom: 3

@14721ms: [TM] Attempting to LOCK 'B' | Eff: 2 | Nom: 2

>>>> TM acquires 'B' and is blocked by TL on 'A'. TL inherits TM's  priority. <<<<

@14721ms: [TM] LOCKED 'B', now trying to LOCK 'A' | Eff: 2 | Nom: 2

>>>> TH will blocked by TM on 'B': <<<<

@14722ms: [TH] Attempting to LOCK 'B'| Eff: 1 | Nom: 1

>>>> TM inherits TH's priority. TL inherits TH's priority via TM. <<<<

@14780ms: [TL] About to UNLOCK 'A' | Eff: 1 | Nom: 3

>>>> Upon unlocking 'A', TL is preempted by TM. It means TL's priority has been
restored, as it is no longer blocking a higher priority task. <<<<


>>>> Now TM acquires 'A' <<<<

@14780ms: [TM] LOCKED 'A' (in CS) | Eff: 1 | Nom: 2
```

```
>>>> After releasing 'A', but before releasing 'B', TM's priority is still '1', as it
is blocking TH while holding 'B'. <<<<

@14780ms: [TM] UNLOCKING 'B' | Eff: 1 | Nom: 2

>>>> Upon unlocking 'B' TM is preempted by TH. (TM's priority has been restored.) <<<<

@14780ms: [TH] LOCKED 'B' (in CS)  | Eff: 1 | Nom: 1

>>> RESULT: even though priority inversion was enforced, tasks leave the nested lock
ordered by their nominal priority. <<<

--->@14780ms: [TH] Exit CS | Eff: 1 | Nom: 1
--->@14780ms: [TM] Exit CS | Eff: 2 | Nom: 2
--->@14780ms: [TL] Exit CS | Eff: 3 | Nom: 3
```

Importantly, the worst-case time is bounded by the time the lowest priority task holds a lock (60 ms in the example: 14720ms → 14780ms).

As for each priority update we check each waiting queue for each mutex a task owns, t he time-complexity is linear `O(owner*mutex)`. But, typically no task ever holds more than a few mutexes. Yet, one should not be encouraged to nest locks if not needed.

5.2.5. Mutexes vs Binary Semaphores

There is (or used to be) a lot of fuss about whether binary semaphores are appropriate to use as locks. As a practical guideline, if all tasks sharing the resource have the same priority, using a binary semaphore *can be appropriate* — because a binary semaphore is considerably faster. It all depends on the case.

The drawback is the lack of ownership: any task can accidentally release the resource. On a large codebase, this can become a real problem. Nonetheless, this is a problem for semaphores in general.

For tasks with different priorities, binary semaphores should never be considered for mutual exclusion unless priority inversion is not a problem (how?).

Counting semaphores initialised as 1 is too risky. Besides the priority inversion, if the count ever increases above 1, mutual exclusion is lost, and multiple tasks can enter the critical section at once.

5.3. Scheduler Lock

Often, we need a task to perform operations without being preempted. A mutex serialises access to a code region but does not prevent a task from being preempted while operating on data. Depending on the case, this can lead to inconsistent data state.

An aggressive way is to disable interrupts globally. For kernel services often it is the only way to keep data integrity. On the higher level it is feasible for very short operations and/or when you need

to protect data from interrupts altogether.

A less aggressive approach is to make the task non-preemptible with `kSchLock()` before entering the critical region and `kSchUnlock()` when leaving. This way, interrupts are still being sensed, and even higher-priority tasks might switch to a ready state, but the running thread will not be preempted.

The priority inversion it potentially causes is bounded. If a higher-priority task is readied while the scheduler is locked, the context switch happens immediately after unlocking.

Note that for locking/unlocking the scheduler the global interrupts will be disabled for the time to increment/decrement a counter, therefore, if your atomic operation is as short as that (3 to 4 cycles), disabling/enabling global interrupts is a better alternative.

To add to the discussion, when two threads need to access the same data to 'read-modify-write', a lock-free mechanism is the LDREX/STREX operations of ARMv7M (or more generally C11 atomics). They do not avoid preemptions, and particularly in ARMv7m, if the data is touched by an ISR before the store-exclusive concludes, the ownership is lost. Typically used for multi-core spin-locking.

5.4. Sleep Queue

Sleep Queue Control Block
Task Waiting Queue

`RK_SLEEP_QUEUE` is a queue of tasks sleeping until explicitly woken.

This primitive was formerly named `RK_EVENT`. Queue names usually reflect the condition they represent (for example, `notFull`, `notEmpty`, `goWriters`, `goReaders`).

An *RK_SLEEP_QUEUE* is stateless: it does not record whether an event has happened. Therefore `kSleepQueueWait(&q, timeout)` always attempts to sleep the caller. `RK_NO_WAIT` is effectively a no-op on this primitive.

`kSleepQueueSignal(&q)` wakes one waiter (highest priority first).

`kSleepQueueWake(&q, n, &u)` wakes at most `n` waiters.

`kSleepQueueFlush(&q)` is the wake-all form (`n = 0`).

`kSleepQueueQuery(&q, &n)` returns the number of waiting tasks.

`kSleepQueueSuspend(&q, taskHandle)` can move a `READY` task to that sleep queue as `SLEEPING_SUSPENDED`. (task context or ISR). Other states are not allowed to be changed.

`kSleepQueueReady(&q, taskHandle)` does the opposite, moving a specific task from that sleep queue to `READY`.

`kSleepQueueWake()` is synchronous in thread context. When invoked from ISR, the wake request is

queued for post-processing by the system task (`uTasksPtr` must be `NULL` in this case).

Because Sleep Queues are stateless, they are prone to lost wake-ups when used alone. They are best used as building blocks for Conditional Critical Regions (monitor-like constructs).

5.5. Conditional Critical Regions

Task Events and semaphores work by atomically updating state and testing predicates that control execution flow (for example, `pend` on a semaphore with count `0` blocks the caller).

A critical region guarded by a lock is either free or taken. What if we need to wait on a richer condition?

We can combine a Mutex and a Sleep Queue, but there is an important detail: operations on both must be coordinated atomically.

When that combination is wrapped as data plus methods, it is a *Monitor*.

5.5.1. Monitor Invariants

1. a single task can be active within a monitor;
2. only the active task within a monitor can check or change its state.

Since tasks signal or sleep *within* a monitor based on its *internal* state, and only one task can be active, we need to establish how to keep both invariants true.

5.5.2. Signalling Discipline

At any moment a single task can be active within a monitor. When the sleeping task is signalled, there are 3 common disciplines to follow: signal-and-leave (Hansen), signal-and-wait (Hoare) or signal-and-continue (Mesa).

Arguably, the most common is *signal-and-continue* - rather than leaving or suspending itself the active task might continue within the monitor. That is possible if the active task holds a lock the waking task needs to acquire to enter. Upon leaving, the active task must release the lock.

The major implication is that by the time the woken task enters the monitor, the condition it was waiting for might no longer be true. It sounds odd because a Monitor is about encapsulating a conditional critical region so no outsiders change its state. But, either a flush, a bad design — or a preemption anomaly — can violate that somehow.

Mesa Monitor has a typical test-loop pattern:

```
--- snippet ---
while (condition.isFalse())
{
```

```
    /*unlock-wait sequence:*/
     DISABLE_PREEMPTION
     unlock(mutex);
     sleep(condition)
     ENABLE_PREEMPTION
     lock(mutex);
     /* when waking, the while clause is tested again */
 }
 --- snippet ---
```

The figure below shows a producer-consumer problem if implemented on a Monitor-idiom.

```
                        Entry Queue [][][]▯>
        ..............................................
        -------------------
        | Internal Data   |
        -------------------  |[Active Task]|
        | Sleep Queues    |
        |Sleeping for Data|
        |Sleeping for Room|
        -------------------
                             -> [][][] Exit Queue
       ...............................................
```

The `UNLOCK-SLEEP` sequence within the testing loop has preemption disabled because after releasing the lock, the task cannot be allowed to resume within the monitor again, for any reason that is not the monitor predicate being satisfied.

5.5.3. Condition Variable Model

The *Condition Variable Model* allows a task to wait within a monitor-construct and if active, operate using `signal()`, `wait()` and `broadcast()` respecting the monitor invariant.

Sleep Queues are actually the canonical *Condition Variable* introduced by Hoare:

> *Note that a condition "variable" is neither true nor false; indeed, it does not have any stored value accessible to the program. In practice, a condition variable will be represented by an (initially empty) queue of processes which are currently waiting on the condition; but this queue is invisible both to waiters and signallers. This design of the condition variable has been deliberately kept as primitive and rudimentary as possible (...)*
>
> (*Monitors: An Operating System Structuring Concept, Hoare, 1974*)

RK0 follows *pthreads condition variable* semantics, aligned with Mesa monitors.

The difference is that there is no standalone *CondVar* primitive. The programmer combines Sleep Queues and Mutexes and uses these helpers:

- `kCondVarWait(&sleepq, &mutex, timeout)`
- `kCondVarSignal(&sleepq)`
- `kCondVarBroadcast(&sleepq)`

The `condWait` is the real helper. When using it, a Mesa testing-loop reduces to:

```
while(!condition)
{
  kCondVarWait(&condQueue, &monitorLock, timeout);
}
```

If you need a monitor policy different from Mesa, you can build it from the same primitives.

> `kCondVarWait`, `kCondVarSignal`, and `kCondVarBroadcast` are task-context APIs and cannot be called from ISRs.

5.5.3.1. Usage Example: Synchronisation Barrier

A given number of tasks must reach a point in the program before *all can proceed*, so every task calls a `barrWait(&barrier)` to catch up with the set of tasks it must synchronise.

The last task entering the barrier will broadcast a signal to all tasks waiting for the wake condition.

At any moment within a Monitor a single task is RUNNING (what is an invariant of the kernel), all other tasks within the monitor are either SLEEPING (for some condition) or BLOCKED (on a mutex).

```
/* Synchronisation Barrier */

typedef struct
{
    RK_MUTEX lock;
    RK_SLEEP_QUEUE allSynch;
    UINT count; /* number of tasks in the barrier */
    UINT round; /* increased every time all tasks synch */
    UINT nRequired; /* number of tasks required */
} Barrier_t;

VOID BarrierInit(Barrier_t *const barPtr, UINT nRequired)
{
    kMutexInit(&barPtr->lock, RK_INHERIT);
    kSleepQueueInit(&barPtr->allSynch);
    barPtr->count = 0;
    barPtr->round = 0;
    barPtr->nRequired = nRequired;

}

VOID BarrierWait(Barrier_t *const barPtr)
```

```
{
    UINT myRound = 0;
    kMutexLock(&barPtr->lock, RK_WAIT_FOREVER);

    /* save round number */
    myRound = barPtr->round;
    /* increase count on this round */
    barPtr->count++;

    if (barPtr->count == barPtr->nRequired)
    {
        /* reset counter, inc round, broadcast to sleeping tasks */
        barPtr->round++;
        barPtr->count = 0;
        kCondVarBroadcast(&barPtr->allSynch);
    }
    else
    {
        /* sequence: a proper wake signal might happen after inc round */
        while ((UINT)(barPtr->round - myRound) == 0U)
        {
            RK_ERR err = kCondVarWait(&barPtr->allSynch, &barPtr->lock,
RK_WAIT_FOREVER);
            K_ASSERT(err==RK_ERR_SUCCESS);
        }
    }
    kMutexUnlock(&barPtr->lock);
}


#define N_REQUIRED 3

Barrier_t syncBarrier;

VOID kApplicationInit(VOID)
{

    K_ASSERT(!kCreateTask(&task1Handle, Task1, RK_NO_ARGS, "Task1", stack1, STACKSIZE,
2, RK_PREEMPT));
    K_ASSERT(!kCreateTask(&task2Handle, Task2, RK_NO_ARGS, "Task2", stack2, STACKSIZE,
3, RK_PREEMPT));
    K_ASSERT(!kCreateTask(&task3Handle, Task3, RK_NO_ARGS, "Task3", stack3, STACKSIZE,
1, RK_PREEMPT));
    BarrierInit(&syncBarrier, N_REQUIRED);
}
VOID Task1(VOID* args)
{
    RK_UNUSEARGS
    while (1)
    {
        kPuts("Task 1 is waiting at the barrier...\r\n");
```

```
        BarrierWait(&syncBarrier);
        kPuts("Task 1 passed the barrier!\r\n");
        kSleep(8);
    }
}

VOID Task2(VOID* args)
{
    RK_UNUSEARGS
    while (1)
    {
        kPuts("Task 2 is waiting at the barrier...\r\n");
        BarrierWait(&syncBarrier);
        kPuts("Task 2 passed the barrier!\r\n");
        kSleep(5);
    }
}

VOID Task3(VOID* args)
{
    RK_UNUSEARGS
    while (1)
    {
        kPuts("Task 3 is waiting at the barrier...\r\n");
        BarrierWait(&syncBarrier);
        kPuts("Task 3 passed the barrier!\r\n");
        kSleep(3);
    }
}
```

```
Task 3 is waiting at the barrier...
Task 2 is waiting at the barrier...
Task 1 is waiting at the barrier...
Task 3 passed the barrier!
Task 1 passed the barrier!
Task 2 passed the barrier!
Task 3 is waiting at the barrier...
Task 2 is waiting at the barrier...
Task 1 is waiting at the barrier...
Task 3 passed the barrier!
Task 1 passed the barrier!
Task 2 passed the barrier!
Task 3 is waiting at the barrier...
Task 2 is waiting at the barrier...
Task 1 is waiting at the barrier...
Task 3 passed the barrier!
Task 1 passed the barrier!
Task 2 passed the barrier!
Task 3 is waiting at the barrier...
Task 2 is waiting at the barrier...
Task 1 is waiting at the barrier...
Task 3 passed the barrier!
Task 1 passed the barrier!
Task 2 passed the barrier!
```

Monitor/CondVars are not to be signalled by ISRs. Because as they do not accumulate and an ISR is not a normal task, it can't be within a monitor and a signal might be lost.

5.5.3.2. Usage Example: Readers Writers Lock

Several readers and writers share a piece of memory. Readers can concurrently access the memory to read; a single writer is allowed (otherwise, data would be corrupted).

When a writer finishes, it checks for any readers waiting. If there is, the writer flushes the readers waiting queue. If not, it wakes a single writer, if any. When the last reader finishes, it signals a writer.

Every read or write operation begins with an acquire and finishes with a release.

*PS: This RWLock implementation has a **reader-preference** policy, as when a writer finishes, it flushes sleeping readers. When the last reader finishes, it will signal writer waiting queue.*

```
/* RW-Lock */

/* a single writer is allowed if there are no readers */
/* several readers are allowed if there is no writer*/
typedef struct
{
    RK_MUTEX     lock;
    RK_SLEEP_QUEUE   writersGo;
    RK_SLEEP_QUEUE   readersGo;
    INT          rwCount; /* number of active readers if > 0 */
                          /* active writer if -1             */

}RwLock_t;

VOID RwLockInit(RwLock_t *const rwLockPtr)
{

    kMutexInit(&rwLockPtr->lock, RK_INHERIT);
    kSleepQueueInit(&rwLockPtr->writersGo);
    kSleepQueueInit(&rwLockPtr->readersGo);
    rwLockPtr->rwCount = 0;
}

/* A writer can acquire if  rwCount = 0 */
/* An active writer is indicated by rwCount = -1; */
VOID RwLockAcquireWrite(RwLock_t *const rwLockPtr)
{
    kMutexLock(&rwLockPtr->lock, RK_WAIT_FOREVER);
    /* if different than 0, there are either writers or readers */
    /* sleep to be signalled */
    while (rwLockPtr->rwCount != 0)
    {
```

```
        kCondVarWait(&rwLockPtr->writersGo, &rwLockPtr->lock, RK_WAIT_FOREVER);
        /* mutex is locked when waking up*/
    }
    /* woke here, set an active writer */
    rwLockPtr->rwCount = -1;
    kMutexUnlock(&rwLockPtr->lock);
}

/* a writer releases, waking up all waiting readers, if any */
/* if there are no readers, a writer can get in */
VOID RwLockReleaseWrite(RwLock_t *const rwLockPtr)
{
    kMutexLock(&rwLockPtr->lock, RK_WAIT_FOREVER);

    rwLockPtr->rwCount = 0; /* indicate no writers*/

    /* if there are waiting readers, flush */
    ULONG nWaitingReaders=0;
    kSleepQueueQuery(&rwLockPtr->readersGo, &nWaitingReaders);
    if (nWaitingReaders > 0)
    {
        /* condVarBroadcast is just an alias for an event flush */
        kCondVarBroadcast(&rwLockPtr->readersGo);
    }
    else
    {
        /* wake up a single writer if any */
        kCondVarSignal(&rwLockPtr->writersGo);
    }
    kMutexUnlock(&rwLockPtr->lock);
}

/* a reader can acquire if there are no writers */
VOID RwLockAcquireRead(RwLock_t *const rwLockPtr)
{
    kMutexLock(&rwLockPtr->lock, RK_WAIT_FOREVER);
    /* if there is an active writer, sleep */
    while (rwLockPtr->rwCount < 0)
    {
        kCondVarWait(&rwLockPtr->readersGo, &rwLockPtr->lock, RK_WAIT_FOREVER);
        /* mutex is locked when waking up*/
    }
    /* increase rwCount, so its > 0, indicating readers */
    rwLockPtr->rwCount ++;
    kMutexUnlock(&rwLockPtr->lock);
}

/* a reader releases and wakes a single writer */
/* if it is the last reader */
VOID RwLockReleaseRead(RwLock_t *const rwLockPtr)
{
```

```
    kMutexLock(&rwLockPtr->lock, RK_WAIT_FOREVER);
    rwLockPtr->rwCount --;
    if (rwLockPtr->rwCount == 0)
    {
        kCondVarSignal(&rwLockPtr->writersGo);
    }
    kMutexUnlock(&rwLockPtr->lock);
}
```

In the image below, 4 tasks — a fast writer (Task 1), a slow writer (Task 4) and two readers (Task3 is faster than Task2) — reading from and writing to a shared UINT variable:

```
15340ms : Task 3 read 73
15390ms : Task 4 wrote 75
15500ms : Task 3 read 75
15550ms : Task 2 read 75
15620ms : Task 1 wrote 76
15750ms : Task 3 read 76
15910ms : Task 2 read 76
16000ms : Task 1 wrote 77
16100ms : Task 3 read 77
16200ms : Task 4 wrote 79
Task4 is flushing readers
16300ms : Task 3 read 79
16350ms : Task 2 read 79
16420ms : Task 1 wrote 80
16520ms : Task 3 read 80
16570ms : Task 4 wrote 82
Task4 is flushing readers
16670ms : Task 2 read 82
16750ms : Task 3 read 82
16800ms : Task 1 wrote 83
```

5.6. Message Queue

The key mechanism for Message Passing in *RK0* is a *Message Queue*. Messages are passed *by copy* and have a fixed-size of 4-,8-,16- or 32-byte, i.e., 1, 2, 4 or 8 words.

Two other abstractions are constructed on top of Queues:

1. *Port*, which is a server endpoint that enqueues *Remote Invocations*, or *Procedure Calls* from clients.
2. *Mailbox*, a 1-word message queue. It can be used as is, and is used by the kernel as reply route from servers to clients.

Message Queue Control Block
Buffer Address
Message Size

Message Queue Control Block
Number of Mesages
Write Position
Read Position
Server Status
Owner Task
Notify callback
Waiting queue

5.6.1. Size of a Message

Each declared queue has a *fixed message-size* at initialisation, and can assume, 1, 2, 4 or 8 WORDs (4, 8, 16, 32 BYTEs). This constraint is intentional. Word-aligned copies are faster, predictable and safer for type casting.

(A word-aligned *single copy* will take ~5 cycles in Cortex-M3/4/7, and ~6 cycles on Cortex-M0/M0+.)

Ports have messages whose types are labeled as:

- `RK_PORT_MESG_<2,4,8>WORD`, which the first two are reserved meta-data. For variable payload one uses the `RK_PORT_MESG_COOKIE`, passing an opaque spointer as payload.

5.6.2. Blocking and non-blocking behaviour

A producer task can optionally block on a full queue, switching its state to `SENDING`. A consumer (optionally) blocks on an empty queue switching its state to `RECEIVING`.

Note that when using non-blocking calls (`RK_NO_WAIT`) they immediately return if unsuccessful.

One can use a `peek()` to read from the head of a queue without extracting the message. A `jam()` method is used to put a message on the queue head, but it does not overwrite. `jam()` is meaningless for Mailboxes.

A `postovw()` overwrites the oldest message and is allowed solely for 1-message queues. If N-message queues were allowed to be overwritten, all unread messages would leak as read and write pointers are adjusted on the operation. In case one needs overwriting full queues continuously a classic ring buffer will do.

`peek()` and `postovw()` are normally used on Mailboxes for last-message semantics — the Mailbox never goes empty; a new message is placed by overwriting the current one.

5.6.3. Ownership and Priority Inversion

Message Queues (and by extension, Mailboxes) *can* have owners, and thus, only the owners can receive at these channels. One might think it makes it a *Port*, but not yet. The fact is, if a receiver is blocked on full queue, the sender can boost its priority.

Ports are for _ synchronous procedure calls_. The server needs to run at the client's priority, which means either boosting or demoting priority.

5.6.4. Notify callback

A callback can be registered for when a queue sends a message successfuly, as a means of event notification.

5.6.5. Usage Examples

Below there is a rich set of usage examples. They demonstrate how to use message-passing in RK0, its API and helper macros to declare storage and messages at the appropriate size.

5.6.5.1. Mail Queue pattern

In RTOS jargon, a *Mailbox* is often regarded as a 1-slot queue. The size can vary but a commonly it is a pointer-sized message `(sizeof(VOID*))`.

When exchanging messages which carry data to be parsed, for instance:

```
DATA STRUCT:
[COMMAND ID]
[PAYLOAD SIZE]
[PAYLOAD]
[CRC]
```

On the *Mail Queue* pattern, a buffer is allocated, filled and sent through the message queue by a producer.

A consumer dequeues, use data (may copy it locally or not) and when done deallocate it.

This *producer-allocates/consumer-deallocates* contract, guarantees data integrity on the reference. On practice, we are *moving* data between threads.

If we are talking of messages bigger than 1-word, without the allocator we might either copy twice sender→queue buffer→receiver or handing off references with no explicit contract for keeping data scope.

A Mail Queue pattern is used on the Application Logger facility that happens to use several non-blocking senders to a single blocking-receiver, aligned to the Active Object design pattern.

```
/* Application Logger pattern */

/* standard log structure */
struct log
{
    RK_TICK t; /* timestamp */
    CHAR s[LOGLEN]; /*formatted string */
    UINT    level; /* level 0=message, 1=fault */
```

```
} K_ALIGN(4);

typedef struct log Log_t;

/* logger mem allocator + mem pool */
static RK_MEM_PARTITION qMem;
static Log_t logBufPool[LOGPOOLSIZ] K_ALIGN(4);

/* backing buffer for the logger queue */
/* (messages are 1-word-size, number equals the pool) */
RK_DECLARE_MESG_QUEUE_BUF(logQBuf, VOID *, LOGPOOLSIZ)

/* logger mail queue */
static RK_MESG_QUEUE logQ;

/* a sender will allocate a buffer write the log
message and enqueue on the mail queue, not-blocking
if the queue is full it returns the buffer immediately
if the memory pool is empty it drops the operation */

```c
/* excerpt of logPost(...) */
VOID logPost(/*formatted string */)
    ---snip---
    Log_t *logPtr = (Log_t*)kMemPartitionAlloc(&qMem);
    RK_BARRIER
    if (logPtr) /* available buffer */
    {

       < fill the buffer >

        /* enqueue address */
        if (kMesgQueueSend(&logQ, &p, RK_NO_WAIT) != RK_ERR_SUCCESS)
        {
            /* queue is full, deallocate buf */
            RK_ERR err = kMemPartitionFree(&qMem, &p);
            K_ASSERT(err==RK_ERR_SUCCESS);
        }

    }
    ---snip---

/* excerpt of the logger task */

static VOID LoggerTask(VOID *args)
{
    RK_UNUSEARGS
    while (1)
    {

        VOID *recvPtr = NULL;
```
```

```
        /* drain the queue: keep receiving while successful */
        while (kMesgQueueRecv(&logQ, &recvPtr, RK_WAIT_FOREVER) == RK_ERR_SUCCESS)
        {
            < print buffer contents >

            /* deallocate */
            RK_ERR err = kMemPartitionFree(&qMem, recvPtr);
            K_ASSERT(err == RK_ERR_SUCCESS);

        }
    }
}
```

This is one of the many ways of using the Mail Queue pattern.

The entire implementation can be seen at `app\logger.c`.

5.6.5.2. Averaging Sensor Values

A task receives measured sensor values from an ISR on a periodic rate. (A Soft Timer emulates the ISR).

Then it enqueues this data to a consumer - that will process the average value for each of 4 sensors.

The inter-task communication is designed as follows:

1. The producer pends on a Mailbox that an ISR posts to. An application timer emulates this ISR.
2. The data extracted from the Mailbox is placed in a queue with the processing task as the consumer.
3. As the producer's priority must be higher than that of the consumer, eventually, the queue will get full.
4. The producer drops the last message when the queue is full and signals the consumer.
5. Now the consumer has a batch of data to work until the next sensor update. It will block (pend on a signal) whenever the queue is empty.

```
/* kPend() and kSignal(t) are macro-helpers using kTaskEventGet/Set */

/* sensor type */
typedef enum
{
    TEMPERATURE=1, HUMIDITY, CO2, FLOW
}SensorType_t;

struct sensorMsg
{
    SensorType_t sensorType;
    ULONG sensorValue;
```

```
};
typedef struct sensorMsg Mesg_t;

#define N_SENSOR    4
#define AVG_WINDOW_SIZE   10 /* 10 samples */

/* the queue */
RK_MESG_QUEUE sensorStream;

/* convenience macro to declare the queue storage */
#define N_MESSAGE 8
RK_DECLARE_MESGQ_BUF(mesgBuf, Mesg_t, N_MESSAGE)

/* timer to mimic isr */
RK_TIMER timerT1;

/* the mailbox the sensor task pends */
RK_MAILBOX sensorBox;

/* callback fwded declaration */
VOID callBackISR(VOID* args);

VOID kApplicationInit( VOID)
{
    K_ASSERT(!kMesgQueueInit(&sensorStream,
                             mesgBuf,
                              RK_MESGQUEUE_MESG_SIZE(Mesg_t), /* set mesg size */
                              N_MESSAGE));

    /* timer @ every 10 ms */
    K_ASSERT(!kTimerInit(&timerT1, 0, 10, callBackISR, NULL, RK_TIMER_RELOAD));
    K_ASSERT(!kMailboxInit(&sensorBox));
}


VOID callBackISR(VOID *args)
{

    static Mesg_t sample = {0};
    static Mesg_t *samplePtr = &sample;
    static UINT sampleErr;

    /* fill sample */
    sample.sensorType = (rand() % 4) + 1;
    switch (sample.sensorType)
    {
        case TEMPERATURE:
            sample.sensorValue = ( ULONG) rand() % 50;
            break;
        case HUMIDITY:
```

```
            sample.sensorValue = ( ULONG) rand() % 100;
            break;
        case CO2:
            sample.sensorValue = ( ULONG) rand() % 1000;
            break;
        case FLOW:
            sample.sensorValue = ( ULONG) rand() % 10;
            break;
        default:
            break;
    }
    /* send address */
    RK_ERR err = kMailboxPost(&sensorBox, &samplePtr, RK_NO_WAIT);
    if (err != RK_ERR_SUCCESS)
        sampleErr ++;

}

/* Producer - higher priority, blocks on mailbox */
VOID Task1(VOID *args)
{
    RK_UNUSEARGS
    Mesg_t *recvSample = NULL;
    while (1)
    {
        /* receive a pointer */
        RK_ERR errmbox = kMailboxPend(&sensorBox, &recvSample,RK_WAIT_FOREVER);
        K_ASSERT( errmbox==RK_ERR_SUCCESS);

        /* enqueue by copy into the stream */
        RK_ERR err = kMesgQueueSend(&sensorStream, recvSample, RK_NO_WAIT);

        K_ASSERT(err >= 0); /* either succesful or unsuccesful */
        if (err == RK_ERR_SUCCESS)
        {
            CHAR const *sensorTypeStr = NULL;
            if (recvSample->sensorType == 1)
                sensorTypeStr = "TEMP";
            if (recvSample->sensorType == 2)
                sensorTypeStr = "HUM";
            if (recvSample->sensorType == 3)
                sensorTypeStr = "CO2";
            if (recvSample->sensorType == 4)
                    sensorTypeStr = "FLOW";
            printf( "ENQ: [@%lums, %s, %lu] \r\n", kTickGet(), sensorTypeStr,
                        recvSample->sensorValue);
        }
        /* full, drop this sample and signal task2 */
        else if (err == RK_ERR_MESGQ_FULL)
        {
            kSignal(task2Handle);
```

```
        }
    }
}

/* for each sensor:
     . a ring buffer of AVG_WINDOW_SIZE values
 . sum of values
 . an index table (=enum - 1 eg., HUMIDITY IDX=2-1=1)
 */
static ULONG ringBuf[N_SENSOR][AVG_WINDOW_SIZE];
static ULONG ringSum[N_SENSOR] = {0};
static UINT ringIndex[N_SENSOR] = {0};

void Task2( void *args)
{

    RK_UNUSEARGS
    Mesg_t readSample;
    while (1)
    {
        RK_ERR err = kMesgQueueRecv(&sensorStream, &readSample, RK_NO_WAIT);
        if (err == RK_ERR_SUCCESS)
        {
            UINT sensorIdx = readSample.sensorType - 1;

/* remove oldest sample */
            ULONG oldest = ringBuf[sensorIdx][ringIndex[sensorIdx]];
            ringSum[sensorIdx] -= oldest;

/* push new sample */
            ringBuf[sensorIdx][ringIndex[sensorIdx]] = readSample.sensorValue;
            ringSum[sensorIdx] += readSample.sensorValue;

/* index incr-wrap */
            ringIndex[sensorIdx] ++;
            ringIndex[sensorIdx] %= AVG_WINDOW_SIZE;

/* simple average */
            ULONG avg = ringSum[sensorIdx] / AVG_WINDOW_SIZE;


            CHAR const *sensorTypeStr = NULL;
            if (readSample.sensorType == 1)
                sensorTypeStr = "TEMP";
            if (readSample.sensorType == 2)
                sensorTypeStr = "HUM";
            if (readSample.sensorType == 3)
                sensorTypeStr = "CO2";
            if (readSample.sensorType == 4)
                sensorTypeStr = "FLOW";
```

```
            printf( "DEQ: [@%lums, %s, %lu] | AVG: %lu \r\n", kTickGet(),
                    sensorTypeStr, readSample.sensorValue, avg);

        }
        else
        {
            kPend(RK_WAIT_FOREVER);
        }

    }
}
```

```
OUTPUT:

ENQ: [@550ms, CO2, 571]
ENQ: [@560ms, FLOW, 4]
ENQ: [@570ms, FLOW, 4]
ENQ: [@580ms, HUM, 25]
ENQ: [@590ms, CO2, 931]
ENQ: [@600ms, CO2, 487]
ENQ: [@610ms, FLOW, 7]
ENQ: [@620ms, HUM, 79]

>>> Queue is full. Now offload and process. Note the order remains <<<

DEQ: [@630ms, CO2, 571] | AVG: 460
DEQ: [@631ms, FLOW, 4] | AVG: 5
DEQ: [@632ms, FLOW, 4] | AVG: 5
DEQ: [@633ms, HUM, 25] | AVG: 52
DEQ: [@634ms, CO2, 931] | AVG: 553
DEQ: [@635ms, CO2, 487] | AVG: 549
DEQ: [@636ms, FLOW, 7] | AVG: 5
DEQ: [@637ms, HUM, 79] | AVG: 55

>>> Consumer is preempted <<<
ENQ: [@640ms, CO2, 913]
ENQ: [@650ms, CO2, 134]
ENQ: [@660ms, HUM, 47]
ENQ: [@670ms, HUM, 30]
ENQ: [@680ms, TEMP, 7]
ENQ: [@690ms, CO2, 726]
ENQ: [@700ms, FLOW, 7]
ENQ: [@710ms, TEMP, 43]

DEQ: [@720ms, CO2, 913] | AVG: 578
DEQ: [@721ms, CO2, 134] | AVG: 543
DEQ: [@722ms, HUM, 47] | AVG: 51
DEQ: [@723ms, HUM, 30] | AVG: 44
DEQ: [@724ms, TEMP, 7] | AVG: 20
DEQ: [@725ms, CO2, 726] | AVG: 592
```

```
DEQ: [@726ms, FLOW, 7] | AVG: 5
DEQ: [@727ms, TEMP, 43] | AVG: 23
```

5.6.5.3. Queue Select using Notify Callback

A task is receiving from many queues and need to know which one has been able to complete.

The `notifyCbk(queue*)` is executed every time a send is successful. In this case it is using an Event Signal to a task. The Signals Flag indicate the queue number - as a contract - of which queue has completed sends. Note that as sends may coalesce, while a flag caps at 1, the consumer will drain each queue until it is empty, or it is preempted.

```
/* Many-to-1 queue channels */

/* Consumer Select queue based on its event
flags, that a succesfull send triggers  */

#define LOG_PRIORITY 4 /* keep logger as lowest-priority user task */
#define STACKSIZE 256

#define NQUEUES 3 /* number of queues */
#define QSIZ 8 /* depth of each queue */

#define Q0_FLAG   RK_EVENT_1 // (1<<0)
#define Q1_FLAG   RK_EVENT_2 // (1<<1)
#define Q2_FLAG   RK_EVENT_3 // (1<<2)
#define QFLAGS   (ULONG)(Q0_FLAG | Q1_FLAG | Q2_FLAG)

typedef struct
{
    RK_TASK_HANDLE producer;
    UINT payload;
} MESG_t;

/* Succesful Send callbacks */
/* each callback follows this pattern */
static inline
VOID sendNotify0(RK_MESG_QUEUE *qPtr)
{
    (VOID)qPtr;
    kTaskEventSet(consumerHandle, Q0_FLAG);
    /* Q1 flag for queue1 and so forth */
}

/* each callback in installed using kMesgQueueInstallSendCbk
on kApplicationInit() */

/* helper to send */
static inline
VOID enqueueSample(RK_MESG_QUEUE *qPtr UINT payload)
```

```
{
    MESG_t mesg = {
        .payload = payload,
        .producer = RK_RUNNING_HANDLE,
    };
    RK_ERR err = kMesgQueueSend(qPtr, &mesg, RK_WAIT_FOREVER);
    K_ASSERT(err == RK_ERR_SUCCESS);
}


VOID Prod0Task(VOID *args)
{
    RK_UNUSEARGS
    UINT seq = 0U;

    while (1)
    {
        enqueueSample(&queues[0], seq++);
        kSleepRelease(25); /* fast producer */
    }
}

VOID Prod1Task(VOID *args)
{
    UINT seq = 0U;
    RK_UNUSEARGS

    while (1)
    {
        enqueueSample(&queues[1], seq++);

        /* every fourth sample, also tickle the third queue */
        if ((seq & 0x3U) == 0U)
        {
            enqueueSample(&queues[2], seq);
        }

        kSleepRelease(60);
    }
}

/* Consumer listens on all queues, selecting those on its signal flags. */
VOID ConsumerTask(VOID *args)
{
    RK_UNUSEARGS

    MESG_t recv = {0};
    ULONG flags = 0UL;

    while (1)
    {
```

```
        flags = 0UL;
        kTaskEventGet(QFLAGS, RK_EVENT_FLAGS_ANY, &flags,
                           RK_WAIT_FOREVER);

        for (UINT i = 0; i < NQUEUES; ++i)
        {
            if (flags & (1UL << i))
            {
                while (kMesgQueueRecv(&queues[i], (VOID*)&recv, RK_NO_WAIT) ==
                       RK_ERR_SUCCESS)
                {
                    logPost("Q%u <- sender=%s payload=%u", i,
                            RK_TASK_NAME(recv.producer), recv.payload);
                }
            }
        }
    }
}
```

```
       0 ms :: Q1 <- sender=Prod1 payload=0
     250 ms :: Q0 <- sender=Prod0 payload=1
     500 ms :: Q0 <- sender=Prod0 payload=2
     600 ms :: Q1 <- sender=Prod1 payload=1
     750 ms :: Q0 <- sender=Prod0 payload=3
    1000 ms :: Q0 <- sender=Prod0 payload=4
    1200 ms :: Q1 <- sender=Prod1 payload=2
    1250 ms :: Q0 <- sender=Prod0 payload=5
    1500 ms :: Q0 <- sender=Prod0 payload=6
    1750 ms :: Q0 <- sender=Prod0 payload=7
    1800 ms :: Q1 <- sender=Prod1 payload=3
    1800 ms :: Q2 <- sender=Prod1 payload=4
    2000 ms :: Q0 <- sender=Prod0 payload=8
    2250 ms :: Q0 <- sender=Prod0 payload=9
    2400 ms :: Q1 <- sender=Prod1 payload=4
    2500 ms :: Q0 <- sender=Prod0 payload=10
    2750 ms :: Q0 <- sender=Prod0 payload=11
    3000 ms :: Q0 <- sender=Prod0 payload=12
    3000 ms :: Q1 <- sender=Prod1 payload=5
    3250 ms :: Q0 <- sender=Prod0 payload=13
    3500 ms :: Q0 <- sender=Prod0 payload=14
    3600 ms :: Q1 <- sender=Prod1 payload=6
    3750 ms :: Q0 <- sender=Prod0 payload=15
    4000 ms :: Q0 <- sender=Prod0 payload=16
    4200 ms :: Q1 <- sender=Prod1 payload=7
    4200 ms :: Q2 <- sender=Prod1 payload=8
    4250 ms :: Q0 <- sender=Prod0 payload=17
    4500 ms :: Q0 <- sender=Prod0 payload=18
    4750 ms :: Q0 <- sender=Prod0 payload=19
    4800 ms :: Q1 <- sender=Prod1 payload=8
    5000 ms :: Q0 <- sender=Prod0 payload=20
```

```
    5250 ms :: Q0 <- sender=Prod0 payload=21
```

5.6.5.4. Synchronous Client-Server Procedure Call

The example below computes a CRC on the server for the client's payload and returns it as the reply code. Logically, this is unbuffered: the client blocks until it receives the reply.

For `kPortSendRecv()`, each client task must pass the address of a Mailbox to store the client response (4-byte).

Note that the server may temporarily adopt the client priority while serving the request. After `kPortReplyDone()`, it returns to nominal priority.

```
#include <application.h>
#include <logger.h>
#define STACKSIZE 256
#define PORT_MSG_WORDS 4U   /* 2 words meta + 2 words payload */
#define PORT_CAPACITY 16

/* tasks */
RK_DECLARE_TASK(serverHandle, ServerTask,    stack1, STACKSIZE)
RK_DECLARE_TASK(clientHandle, ClientTask,    stack2, STACKSIZE)

/* port */
static RK_PORT serverPort;
RK_DECLARE_PORT_BUF(portBuf, PORT_MSG_WORDS, PORT_CAPACITY)

/* 4-word message format; first two words are metadata */
typedef RK_PORT_MESG_4WORD RpcMsg;

static inline UINT crc32(const VOID *data, ULONG size);
static inline BYTE xorshift8(void);

VOID kApplicationInit(void)
{
    K_ASSERT(!kCreateTask(&serverHandle, ServerTask, RK_NO_ARGS, "Server", stack1,
STACKSIZE, 1, RK_PREEMPT));
    K_ASSERT(!kCreateTask(&clientHandle, ClientTask, RK_NO_ARGS, "Client", stack2,
STACKSIZE, 2, RK_PREEMPT));

    /* init port */
    K_ASSERT(!kPortInit(&serverPort, portBuf, PORT_MSG_WORDS, PORT_CAPACITY,
serverHandle));

    logInit();
}

VOID ServerTask(VOID *args)
{
    RK_UNUSEARGS
```

```
    RpcMsg msg;
    while(1)
    {
        /* receive next request; server may adopt client priority here */
        K_ASSERT(!kPortRecv(&serverPort, &msg, RK_WAIT_FOREVER));

        BYTE  *vector = (BYTE*) msg.payload[0];
        ULONG   size  =          msg.payload[1];
        UINT    crc   = crc32(vector, size);

        logPost("[SERVER] Will Reply CRC=0x%04X | Eff Prio=%d | Nom Prio=%d",
                crc, runPtr->priority, runPtr->prioNominal);

        /* must end with kPortReplyDone */
        K_ASSERT(!kPortReplyDone(&serverPort, (ULONG const*)&msg, crc));

        logPost("[SERVER] Finished. | Eff Prio: %d | Nom Prio: %d", runPtr->priority,
runPtr->prioNominal);
    }
}

static RK_MAILBOX replyBox;

VOID ClientTask(VOID *args)
{
    RK_UNUSEARGS
    static BYTE vec[8];
    for (UINT i = 0; i < 8; ++i)
        vec[i] = xorshift8();

    RpcMsg msg = {0};
    msg.payload[0] = (ULONG) vec;  /* pointer as one word */
    msg.payload[1] = 8;            /* number of bytes */

    UINT reply = 0;
    while(1)
    {
        /* Send-Receive: a call */
        UINT want = crc32(vec, 8);
        K_ASSERT(!kPortSendRecv(&serverPort, (ULONG*)&msg, &replyBox  &reply,
RK_WAIT_FOREVER));
        logPost("[CLIENT] Need=0x%04X | Recvd=0x%04X", want, reply);
        /* if reply is correct, generate a new payload */
        if (want == reply)
            for (UINT i = 0; i < 8; ++i) vec[i] = xorshift8();
        kSleepRelease(1000);
    }
}
```

```
   28000 ms :: [CLIENT] Need=0xC2A6C337 | Recvd=0xC2A6C337
```

```
29000 ms :: [SERVER] Will Reply CRC=0x93F4110A | Eff Prio=2 | Nom Prio=1
29000 ms :: [SERVER] Finished. | Eff Prio: 1 | Nom Prio: 1
29000 ms :: [CLIENT] Need=0x93F4110A | Recvd=0x93F4110A
30000 ms :: [SERVER] Will Reply CRC=0x7A8FA006 | Eff Prio=2 | Nom Prio=1
30000 ms :: [SERVER] Finished. | Eff Prio: 1 | Nom Prio: 1
30000 ms :: [CLIENT] Need=0x7A8FA006 | Recvd=0x7A8FA006
31000 ms :: [SERVER] Will Reply CRC=0x9051C8B1 | Eff Prio=2 | Nom Prio=1
31000 ms :: [SERVER] Finished. | Eff Prio: 1 | Nom Prio: 1
31000 ms :: [CLIENT] Need=0x9051C8B1 | Recvd=0x9051C8B1
32000 ms :: [SERVER] Will Reply CRC=0x11F29117 | Eff Prio=2 | Nom Prio=1
32000 ms :: [SERVER] Finished. | Eff Prio: 1 | Nom Prio: 1
```

5.7. Most-Recent Message Protocol (MRM)

MRM Control Block
MRM Buffer Allocator
Data Buffer Allocator
Current MRM Buffer Address
Data Size (Message Size)

MRM Buffer
Data Buffer Address
Readers Count

Data Buffer
Application-dependent

> There is little practical difference between a message that does not arrive and one with no valid (stale) data. But when wrong (or stale) data is processed - e.g., to define a set point on a loop - a system can fail badly.
>
> Design Choice: provide a broadcast asynchronous message-passing scheme that guarantees data freshness and integrity for all readers.
>
> Benefits: The system has a mechanism to meet strict deadlines that cannot be predicted on design time.

Control loops reacting to unpredictable time events—like a robot scanning an environment or a drive-by-wire system—require a different message-passing approach. Readers cannot "look at the past" and cannot block. The most recent data must be delivered lock-free and have guaranteed integrity.

As *Ports*, the *MRM* is a high-level mechanism. It was chosen to be provided as a kernel service, given its distinctive nature and suitability for *RK0* applications.

5.7.1. Functional Description

An *MRM* works as a *1-to-many asynchronous Mailbox* - with a lock-free specialisation that enables several readers to get the most recent deposited message with no integrity issues. Whenever a reader reads an MRM buffer, it will find the most recent data transmitted. It can also be seen as an extension of the Double Buffer pattern for a 1:N communication.

The core idea of the MRM protocol is that readers can only access the buffer that is classified as the '*most recent buffer*'. After a writer *publish()* a message, that will be the only message readers can *get()* — any former message being processed by a reader was grabbed *before* a new *publish()* - and, from now on, can only be *unget()*, eventually returning to the pool.

To clarify further, the communication steps are listed:

1. A producer first reserves an MRM Buffer - the reserved MRM Buffer is not available for reading until it is published.
2. A message buffer is allocated and filled, and its address is within an MRM Buffer. The producer *publishes* the message. From now on, it is *the most recent message*. Any former published buffer is no longer visible to new readers
3. A reader starts by *getting* an MRM Buffer. A `get()` operation delivers a copy of the message to the reader's scope. Importantly, this operation increases the number of readers associated to that MRM Buffer.

Before ending its cycle, the task releases (`unget()`) the buffer; on releasing, the kernel checks if the caller task is the last reader and if the buffer being released is not the current MRM Buffer.

If the above conditions are met, the `unget()` operation will return the MRM buffer to the pool. If there are more readers, OR if it is the current buffer, it remains available.

When the `reserve` operation detects that the most recent buffer still has readers, a new buffer is allocated to be written and published. If it has no readers, it is reused.

This way, the worst case is a sequence of `publish()` with no `unget()` at all — this would lead to the writer finding no buffer to reserve. This is prevented by making: `N Buffers = N tasks + 1`.

5.7.2. MRM Control Block Configuration

What might lead to some confusion when initialising an MRM Control Block is the need for two different pools:

- One pool will be the storage for the MRM Buffers, which is the data structure for the mechanism.
- Another pool is for the actual payload. The messages.

Both pools must have the same number of elements: the number of tasks communicating + 1.

- The size of the data buffers is application-dependent - and is passed as a number of *words*. The minimal message size is 32-bit.
- If using data structures, keep it aligned to 4 to take advantage of the performance of aligned

memory.

5.7.3. Usage Example

Consider a modern car - speed variations are of interest in many modules. With a somehow "naive" approach, let us consider three modules and how they should react when speed varies:

1. **Cruiser Control:** For the Cruiser Control, a speed increase might signify the driver wants manual control back, and it will likely turn off.
2. **Windshield Wipers:** If they are on, a speed change can affect the electric motor's adjustments to the air resistance.
3. **Radio:** Speed changes reflect the aerodynamic noise - the radio volume might need adjustment.

As the variations are unpredictable, we need a mechanism to deliver the last speed in order of importance for all these modules. From highest to lowest priority, Cruise, Wipers, and Radio are the three modules that range from safety to comfort.

To emulate this scenario, we can write an application with a higher priority task that sleeps and wakes up at pseudo-random times to produce random values that represent the (unpredictable) speed changes.

The snippet below has 4 periodic tasks. Tasks are periodic using the `kSleepRelease()` primitive.

There is a sequential counter. When a task wakes and see it was not incremented it just runs.

The producer publishes new data at a random interval, preempting whatever task is running at the moment. Given tha random updates,

```
typedef struct
{
    UINT speed;
    ULONG timeStamp;
} Mesg_t;

#define STACKSIZE 256
#define N_MRM (5)                          /* Number of MRMs N Tasks + 1 */
#define MRM_MESG_SIZE (sizeof(Mesg_t) / 4) /* In WORDS */
RK_MRM MRMCtl;                             /* MRM control block */
RK_MRM_BUF buf[N_MRM];                     /* MRM pool */
Mesg_t data[N_MRM];                        /* message data pool */

RK_DECLARE_TASK(speedSensorHandle, SpeedSensorTask, stack1, STACKSIZE)
RK_DECLARE_TASK(cruiserHandle, CruiserTask, stack2, STACKSIZE)
RK_DECLARE_TASK(wiperHandle, WiperTask, stack3, STACKSIZE)
RK_DECLARE_TASK(radioHandle, RadioTask, stack4, STACKSIZE)
volatile UINT seq = 0;
VOID kApplicationInit(VOID)
{
```

```
    kCreateTask(&speedSensorHandle, SpeedSensorTask, RK_NO_ARGS, "SpeedTsk",
                stack1, STACKSIZE, 1, RK_PREEMPT);

    kCreateTask(&cruiserHandle, CruiserTask, RK_NO_ARGS, "CruiserTsk", stack2,
                STACKSIZE, 2, RK_PREEMPT);

    kCreateTask(&wiperHandle, WiperTask, RK_NO_ARGS, "WiperTsk", stack3,
                STACKSIZE, 3, RK_PREEMPT);

    kCreateTask(&radioHandle, RadioTask, RK_NO_ARGS, "RadioTsk", stack4,
                STACKSIZE, 4, RK_PREEMPT);

    kMRMInit(&MRMCtl, buf, data, N_MRM, MRM_MESG_SIZE);

    logInit(5);
}

VOID SpeedSensorTask(VOID *args)
{
    RK_UNUSEARGS

    Mesg_t sendMesg = {0};
    while (1)
    {
        RK_TICK sleepTicks = ((RK_TICK)rand() % 18) + 1;
        kSleep(sleepTicks);
        RK_TICK currTick = kTickGetMs();
        UINT speedValue = (UINT)(rand() % 170) + 1;
        sendMesg.speed = speedValue;
        sendMesg.timeStamp = currTick;
        /* grab a buffer */
        RK_MRM_BUF *bufPtr = kMRMReserve(&MRMCtl);
        if (bufPtr != NULL)
        {
            K_ASSERT(!kMRMPublish(&MRMCtl, bufPtr, &sendMesg));
            printf("!!!!! @%lums SPEED UPDATE: %u mph\r\n", kTickGetMs(),
speedValue);
            seq += 1;
        }
        else
        { /* cannot fail */
            logError("MRM protocol could not find a free buffer\r\n");
        }
        /* publish  */
    }
}

VOID CruiserTask(VOID *args)
{
    RK_UNUSEARGS
    Mesg_t recvMesg = {0};
```

```
    while (1)
    {
        RK_MRM_BUF *readBufPtr = kMRMGet(&MRMCtl, &recvMesg);
        if (readBufPtr)
        {
            logPost("CRUISER: (%u mph, %lu ms)", recvMesg.speed,
recvMesg.timeStamp);

            kMRMUnget(&MRMCtl, readBufPtr);
        }
        kSleepRelease(4);
    }
}

VOID WiperTask(VOID *args)
{
    RK_UNUSEARGS
    Mesg_t recvMesg = {0};

    while (1)
    {

        RK_MRM_BUF *readBufPtr = kMRMGet(&MRMCtl, &recvMesg);
        if (readBufPtr)
        {
            logPost("WIPERS: (%u mph, %lu ms)", recvMesg.speed,
recvMesg.timeStamp);

            kMRMUnget(&MRMCtl, readBufPtr);
        }
        kSleepRelease(7);
    }
}
VOID RadioTask(VOID *args)
{
    RK_UNUSEARGS
    Mesg_t recvMesg = {0};
    while (1)

    {

        RK_MRM_BUF *readBufPtr = kMRMGet(&MRMCtl, &recvMesg);

        if (readBufPtr)
        {
            logPost("RADIO: (%u mph, %lu ms) ", recvMesg.speed,
recvMesg.timeStamp);
            kMRMUnget(&MRMCtl, readBufPtr);
        }
        kSleepRelease(11);
    }
```

```
}
```

Thus, different situations can happen:

- All tasks read the updated pair (speed, time)
- Not all tasks receive the updated pair because another update happens in between.
- No tasks receive an update - because another happens too soon.
- No update happens between in the period of a given task. It receives the same value. No problems.

All these cases are on the log:

```
Logs show: (last speed record, record time)

  !!!@120ms SPEED UPDATE: 164 mph
     120 ms :: CRUISER: (164 mph, 120 ms)
     140 ms :: WIPERS: (164 mph, 120 ms)
 !!! @150ms SPEED UPDATE: 80 mph
     160 ms :: CRUISER: (80 mph, 150 ms)
     200 ms :: CRUISER: (80 mph, 150 ms)
     210 ms :: WIPERS: (80 mph, 150 ms)
     220 ms :: RADIO: (80 mph, 150 ms)
     240 ms :: CRUISER: (80 mph, 150 ms)
  !!!@280ms SPEED UPDATE: 49 mph
     280 ms :: CRUISER: (49 mph, 280 ms)
     280 ms :: WIPERS: (49 mph, 280 ms)
     320 ms :: CRUISER: (49 mph, 280 ms)
     330 ms :: RADIO: (49 mph, 280 ms)
     350 ms :: WIPERS: (49 mph, 280 ms)
     360 ms :: CRUISER: (49 mph, 280 ms)
     400 ms :: CRUISER: (49 mph, 280 ms)
     420 ms :: WIPERS: (49 mph, 280 ms)
     440 ms :: CRUISER: (49 mph, 280 ms)
     440 ms :: RADIO: (49 mph, 280 ms)
  !!!@450ms SPEED UPDATE: 87 mph
     480 ms :: CRUISER: (87 mph, 450 ms)
     490 ms :: WIPERS: (87 mph, 450 ms)
     520 ms :: CRUISER: (87 mph, 450 ms)
 !!! @540ms SPEED UPDATE: 110 mph
     550 ms :: RADIO: (110 mph, 540 ms)
     560 ms :: CRUISER: (110 mph, 540 ms)
     560 ms :: WIPERS: (110 mph, 540 ms)
     600 ms :: CRUISER: (110 mph, 540 ms)
     630 ms :: WIPERS: (110 mph, 540 ms)
 !!! @640ms SPEED UPDATE: 22 mph
     640 ms :: CRUISER: (22 mph, 640 ms)
     660 ms :: RADIO: (22 mph, 640 ms)
```

The highlight is that controllers can keep their pace, while receiving fresh data - you can see it on the timestamp on the image.

Again, they might receive the same data more than once or miss samples; what is important is that *they are not lagging and consuming stale data.*

Chapter 6. *Error Handling*

6.1. Fail fast

While tracing and error handling are yet to be largely improved (and that is when the 1.0.0 version will be released), currently *RK0* employs a policy of *failing fast* in **debug mode**.

When Error Checking is enabled, every kernel call will be 'defensive', checking for correctness of parameters and invariants, null dereferences, etc.

In these cases is more useful to allow the first error to halt the execution by calling an Error Handler function to observe the program state.

A trace structure records the address of the running TCB, its current stack pointer, the link register (that is, the PC at kErrHandler was called), and a time stamp.

This record is on a `.noinit` RAM section, so it is visible if CPU resets. A fault code is stored in a global `faultID` and on the trace structure. Developers can hook in custom behaviour.

If the kernel is configured to not halt on a fault, but Error Checking is enabled, functions will return negative values in case of an error.

On the other hand, when Error Checking is disabled or `NDEBUG` is defined nothing is checked, reducing code size and improving performance.

(*Some deeper internal calls have assertion. For those, only* `NDEBUG` *defined ensures they are disabled.*)

6.2. Stack Overflow

Stack overflow is detected (not prevented) using a "stack painting" with a sentinel word. Stack Overflow detection is enabled by defining the assembler preprocessor `__KDEF_STACKOVFLW` when compiling.

As a matter of fact, sizing your stack is something you must do diligently when programming a system. I would say a mechanism for stack overflow detection is on the bottom of the list of 'must-have' features.

One can take advantage of the static task model - *it is possible to predict offline* the deepest call within any task. The compiler flag `-fstack-usage` creates `.su` files indicating the depth of every function within a module. This is an example of compilation-unit output:

```
core/src/ksema.c:34:8:kSemaphoreInit    88  static
core/src/ksema.c:74:8:kSemaphorePend    96  static
core/src/ksema.c:189:8:kSemaphorePost   88  static
core/src/ksema.c:265:8:kSemaphoreFlush  72  static
core/src/ksema.c:306:5:kSemaphoreQuery  40  static
core/src/kmutex.c:128:8:kMutexInit  16  static
core/src/kmutex.c:165:8:kMutexLock  120 static
```

```
core/src/kmutex.c:325:8:kMutexUnlock    96  static
core/src/kmutex.c:425:6:kMutexQuery 56  static
```

These are the worst cases. Now, you identify the depth of the longest *chain of calls* for a task using these services and add a generous safety margin — 30%. The cap depends on your budget.

Importantly, you also have to size the System Stack. This initial size is defined in `linker.ld` by the symbol `Min_Stack_Size`. In this case, account for the depth of `main()`, `kApplicationInit()`, and all interrupt handlers; again, inspect the longest call chain depth. Assume interrupts always add to the worst static depth, and account for *nested interrupts*.

6.3. Deadlocks

There are deadlock-recovery techniques in the literature, but they are generally unfeasible here. The kernel provides bounded waiting, enforces priority-ordered waiting queues, applies mutex priority inheritance, and offers lock-free primitives and period-drift compensation. None of these techniques prevents deadlocks by itself (and with bounded blocking plus lock-free primitives, one can still get *livelocks*).

- Ordered Locking:

For those programming the application, despite following the RMS rule of higher priority for higher request rate tasks, the golden rule for locking is acquiring resources in an unidirectional order *throughout the entire application*:

```
acquire(A);
acquire(B);
acquire(C);
   .
   .
   .
release(C);
release(B);
release(A);
```

This breaks circular waiting.

For instance:

```
TaskA:
   wait(R1);
   wait(R2);
    /* critical section */
   signal(R2);
   signal(R1);

TaskB:
   wait(R1);
```

```
    wait(R2);
     /* critical section */
    signal(R2);
    signal(R1);
```

But, if:

```
TaskA:
    wait(R1);
    wait(R2);
    .
    .

TaskB:
    wait(R2);
    wait(R1);
    .
    .
```

There are some possible outcomes:

1. Deadlock:
    - TaskA runs: acquires R1
    - TaskB runs: acquires R2
    - TaskA runs: tries to acquire R2 — blocked
    - TaskB runs: tries to acquire R1 — blocked
2. No deadlock:
    - TaskA runs: acquires R1
    - TaskA runs: acquires R2 (nobody is holding R2)
    - TaskA releases both; TaskB runs and acquires both (in either order)

Overall, there is no deadlock if tasks do not overlap in critical sections. That is why systems run for years without deadlocks and eventually: *ploft*.

- Use a 'master-lock' with 'try' semantics:

Another technique that can be employed is if one needs to acquire multiple locks—acquire them all or none using a try-lock (`RK_NO_WAIT`). If any of the tries fail, the task gives up on acquiring the resources and backs off, releasing all successful locks to retry later (most simply, using a sleep queue). That is easier said than done, though, and, as mentioned, if not well done, instead of deadlocks, one gets livelocks.

(*Livelocks* are when a couple of tasks keep running, but the program does not advance.)

Chapter 7. *RK0 Services API*

7.1. Convention

- A kernel call starts with a lowercase `k`. Typically it is followed by a kernel object identifier and an action.

```
kSemaphorePend(&sema, 800); /* pend on a semaphore; 800 ticks time-out */
```

- When `k` is followed by an action, it is acting on the caller task.

```
kSleep(150); /* sleep the caller task for 150 ticks */
```

- Some calls can act either on the caller or on another task:

```
/* stores the signal flags of the task identified by task1Handle on queryValue */
kTaskEventQuery(task1Handle, &queryValue);

/* retrieves its own signal flags */
kTaskEventQuery(NULL, &queryValue);
```

7.2. Return Values

With a few exceptions, kernel calls return a `RK_ERR` error code. `0` is a successful operation `(RK_ERR_SUCCESS)` and any negative value is an error that indicates failure. A positive value is an unsuccessful operation, but will not lead the system to failure (e.g., any unsuccessful `try` operation).

`kapi.h`: RK0 API

Chapter 8. QEMU Unit-Test Invariant Traceability

Invariant ID format: `<MECHANISM_ACRONYM>_RQ_<NN>`.

8.1. Invariants by Mechanism

8.1.1. Scheduler (SCH)

Invariant ID	Invariant	Specification basis	Covered test case(s)
SCH_RQ_01	The scheduler dispatches the highest-priority READY task when a higher-priority task becomes READY.	Task Scheduler	test_kSleepDelay_switches_to_highest_ready_task

8.1.2. Memory Partition (MEM)

Invariant ID	Invariant	Specification basis	Covered test case(s)
MEM_RQ_01	Initialisation rounds block size to 4-byte alignment and sets control-block metadata.	Memory Allocator	test_kMemPartitionInit_aligns_and_initialises_metadata
MEM_RQ_02	Initialisation rejects a NULL partition object and records a fault.	Memory Allocator	test_kMemPartitionInit_rejects_null_object_and_records_fault
MEM_RQ_03	Initialisation is single-shot; double init is rejected.	Memory Allocator	test_kMemPartitionInit_prevents_double_initialisation
MEM_RQ_04	Allocation consumes free blocks in free-list order until exhaustion.	Memory Allocator	test_kMemPartitionAlloc_and_free_manage_free_list_correctly
MEM_RQ_05	`free()` returns blocks to the free list and rejects invalid double-free.	Memory Allocator	test_kMemPartitionAlloc_and_free_manage_free_list_correctly
MEM_RQ_06	Free-list policy is LIFO: most recently freed block is reallocated first.	Memory Allocator	test_kMemPartitionFree_reuses_most_recently_freed_block_first
MEM_RQ_07	Allocation from an invalid object is rejected and raises fault metadata.	Memory Allocator	test_kMemPartitionAlloc_detects_invalid_object

8.1.3. Semaphore (SEMA)

Invariant ID	Invariant	Specification basis	Covered test case(s)
SEMA_RQ_01	Init sets object identity, count bounds, and empty waiting queue.	Semaphore	test_kSemaphoreInit_sets_expected_fields

Invariant ID	Invariant	Specification basis	Covered test case(s)
SEMA_RQ_02	Init rejects invalid inputs (NULL, double init, init > max).	Semaphore	test_kSemaphoreInit_validates_inputs_and_sets_faults
SEMA_RQ_03	pend() decrements count immediately when count is available (S > 0).	Semaphore (P(S) semantics)	test_kSemaphorePend_consumes_available_count
SEMA_RQ_04	pend(RK_NO_WAIT) on zero count returns blocked result without sleeping.	Semaphore non-blocking behaviour	test_kSemaphorePend_respects_no_wait_and_isr_rules
SEMA_RQ_05	Blocking pend() is invalid from ISR context.	Semaphore ISR constraints	test_kSemaphorePend_respects_no_wait_and_isr_rules
SEMA_RQ_06	Timed pend() can time out and clears timeout state for continued execution.	Semaphore bounded waiting	test_kSemaphorePend_can_time_out
SEMA_RQ_07	post() services waiting tasks before counter increments.	Semaphore (V(S) semantics)	test_kSemaphorePost_wakes_highest_priority_waiter, test_kSemaphorePost_waiters_take_precedence_over_counter_increment
SEMA_RQ_08	Binary semaphore saturates at 1 and reports full on further post().	Binary semaphore	test_kSemaphorePost_binary_saturates_without_waiters
SEMA_RQ_09	Counting semaphore saturates at maxValue and reports full on further post().	Counting semaphore	test_kSemaphorePost_counting_saturates_at_max_without_waiters
SEMA_RQ_10	Waiting queue follows priority discipline (highest-priority waiter released first).	Waiting queue priority discipline	test_kSemaphorePost_wakes_highest_priority_waiter, test_kSemaphoreFlush_schedules_highest_priority_blocked_task
SEMA_RQ_11	flush() is broadcast: all waiters released and semaphore count remains 0.	Flushing or Broadcast	test_kSemaphoreFlush_broadcast_releases_all_waiters_and_clears_timeouts
SEMA_RQ_12	flush() on empty waiting queue returns empty-queue status and leaves count unchanged.	Flushing or Broadcast	test_kSemaphoreFlush_returns_empty_waiting_queue_when_nobody_waits
SEMA_RQ_13	flush() from ISR is deferred via post-processing.	Flushing from ISR	test_kSemaphoreFlush_from_isr_uses_deferred_path_and_releases_waiters
SEMA_RQ_14	Invalid timeout values are rejected and caller is not enqueued.	Semaphore timeout validation	test_kSemaphorePend_rejects_invalid_timeout

Invariant ID	Invariant	Specification basis	Covered test case(s)
SEMA_RQ_15	query() returns count when non-negative, or -N where N is number of waiters.	Semaphore query semantics	test_kSemaphoreQuery_reports_waiters_or_count, test_kSemaphoreQuery_reports_negative_waiter_count_magnitude
SEMA_RQ_16	Operations reject invalid semaphore objects.	Semaphore object validity	test_kSemaphorePost_rejects_invalid_objects

8.1.4. Mutex (MUTX)

Invariant ID	Invariant	Specification basis	Covered test case(s)
MUTX_RQ_01	Init accepts only RK_NO_INHERIT or RK_INHERIT and initializes lock state/metadata.	Mutex control block	test_kMutexInit_accepts_valid_protocol_and_initialises_fields, test_kMutexInit_rejects_invalid_protocol_flags
MUTX_RQ_02	Locking a locked mutex with RK_NO_WAIT returns RK_ERR_MUTEX_LOCKED.	Mutex lock semantics	test_kMutexLock_nowait_reports_mutex_locked
MUTX_RQ_03	Mutexes are non-recursive: owner re-lock is rejected.	Mutex non-recursive rule	test_kMutexLock_rejects_recursive_lock
MUTX_RQ_04	Unlocking an unlocked mutex is invalid and rejected.	Mutex unlock validity	test_kMutexUnlock_rejects_unlock_when_mutex_not_locked
MUTX_RQ_05	Unlock by non-owner is invalid and rejected.	Mutex ownership	test_kMutexUnlock_rejects_non_owner
MUTX_RQ_06	Invalid lock timeout values are rejected without queue corruption.	Mutex timeout validation	test_kMutexLock_rejects_invalid_timeout_when_contended
MUTX_RQ_07	Priority inheritance is transitive through lock dependency chains.	Priority Inversion and PIP	test_kMutexLock_applies_transitive_priority_inheritance
MUTX_RQ_08	With RK_NO_INHERIT, owner effective priority is not raised by contention.	Protocol flag semantics	test_kMutexLock_without_inheritance_does_not_raise_owner_priority
MUTX_RQ_09	Unlock handover transfers ownership to a waiting task and restores previous owner nominal priority.	Mutex unlock and PIP restoration	test_kMutexUnlock_handover_transfers_ownership_and_restores_priority
MUTX_RQ_10	query() reports lock state (0 unlocked, 1 locked).	Mutex query	test_kMutexQuery_reports_lock_state

8.1.5. Message Queue (MSGQ)

Invariant ID	Invariant	Specification basis	Covered test case(s)
MSGQ_RQ_01	Queue message size is fixed at init and must be one of 1/2/4/8 words.	Message Queue size of a message	test_kMesgQueueInit_validates_supported_message_sizes
MSGQ_RQ_02	Messages are transmitted by copy.	Message Queue by-copy semantics	test_kMesgQueueSendRecv_is_by_copy_and_preserves_fifo_order
MSGQ_RQ_03	Queue preserves FIFO order.	Message Queue FIFO behaviour	test_kMesgQueueSendRecv_is_by_copy_and_preserves_fifo_order
MSGQ_RQ_04	Non-blocking receive on empty queue returns RK_ERR_MESGQ_EMPTY.	Blocking and non-blocking behaviour	test_kMesgQueue_no_wait_returns_empty_and_full_without_blocking
MSGQ_RQ_05	Non-blocking send on full queue returns RK_ERR_MESGQ_FULL.	Blocking and non-blocking behaviour	test_kMesgQueue_no_wait_returns_empty_and_full_without_blocking
MSGQ_RQ_06	If queue owner is set, only owner task can receive.	Ownership and priority inversion	test_kMesgQueue_owner_only_receiver_is_enforced
MSGQ_RQ_07	Registered send-notify callback runs on each successful send.	Notify callback	test_kMesgQueue_notify_callback_runs_on_successful_send
MSGQ_RQ_08	reset() restores queue to initial state and releases blocked waiters.	Message Queue reset semantics	test_kMesgQueueReset_clears_queue_and_wakes_waiter
MSGQ_RQ_09	reset() from ISR is deferred; immediate queue state remains unchanged until post-processing.	Deferred ISR processing model	test_kMesgQueueReset_in_isr_defers_when_waiters_exist

8.1.6. Port (PORT)

Invariant ID	Invariant	Specification basis	Covered test case(s)
PORT_RQ_01	Port init requires message size including metadata words and binds a server endpoint owner.	Port message format and endpoint ownership	test_kPortInit_requires_metadata_words_and_sets_server_endpoint
PORT_RQ_02	Port owner assignment is single-shot (cannot be reassigned after init).	Port ownership	test_kPortOwner_is_single_assignment

Invariant ID	Invariant	Specification basis	Covered test case(s)
PORT_RQ_03	kPortSend() records sender metadata and clears reply route for one-way request.	Port request metadata	test_kPortSend_writes_sender_metadata_and_null_reply_route
PORT_RQ_04	Only owner task can receive from the port queue.	Port owner-only receive	test_kPortRecv_rejects_non_owner_receiver
PORT_RQ_05	On receive, server may temporarily adopt client priority while serving request.	Synchronous client-server procedure call	test_kPortRecv_and_server_done_move_ready_server_between_priority_queues, test_kPortSend_and_recv_adopts_sender_priority, test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion
PORT_RQ_06	kPortServerDone() restores server priority to nominal.	Server completion semantics	test_kPortRecv_and_server_done_move_ready_server_between_priority_queues, test_kPortSend_and_recv_adopts_sender_priority, test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion
PORT_RQ_07	Priority adoption updates ready-queue placement to prevent unrelated-task inversion.	Priority-aware server execution	test_kPortRecv_priority_adoption_prevents_unrelated_task_inversion, test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion
PORT_RQ_08	Across many clients, dequeue/adoption/restoration invariants remain stable.	Port scheduling invariants	test_kPort_multiple_clients_priority_range_preserves_invariants
PORT_RQ_09	kPortSendRecv() initializes client reply mailbox and writes call metadata (senderHandle, replyBox).	kPortSendRecv() request/reply contract	test_kPortSendRecv_initializes_reply_box_and_sets_call_metadata, test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion
PORT_RQ_10	kPortSendRecv() rejects reply mailbox owned by another task and preserves original message metadata on failure.	Reply-route ownership rule	test_kPortSendRecv_rejects_reply_box_owned_by_other_task
PORT_RQ_11	kPortReply() requires reply-route metadata (replyBox) in message header.	Reply path contract	test_kPortReply_requires_reply_box_in_message_metadata
PORT_RQ_12	kPortReplyDone() demotes server priority even when reply posting fails.	Server demotion rule	test_kPortReplyDone_demotes_server_even_when_reply_fails
PORT_RQ_13	kPortReplyDone() posts reply code into client mailbox and demotes server to nominal priority.	Request/reply completion	test_kPortReplyDone_posts_reply_and_demotes_server, test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion

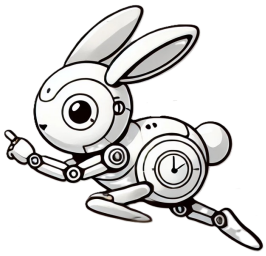
Invariant ID	Invariant	Specification basis	Covered test case(s)
PORT_RQ_14	End-to-end kPortSendRecv() with multiple clients returns per-request replies correctly.	Synchronous procedure call correctness	test_kPortSendRecv_multiple_clients_sum_reply

8.2. Test Case to Invariant Mapping

Module	Test case	REQ ID(s)	Source file
kscheduler	test_kSleepDelay_switches_to_highest_ready_task	SCH_RQ_01	tests/qemu/apps/ut_kscheduler.c
kmem	test_kMemPartitionInit_aligns_and_initialises_metadata	MEM_RQ_01	tests/qemu/apps/ut_kmem.c
kmem	test_kMemPartitionInit_rejects_null_object_and_records_fault	MEM_RQ_02	tests/qemu/apps/ut_kmem.c
kmem	test_kMemPartitionInit_prevents_double_initialisation	MEM_RQ_03	tests/qemu/apps/ut_kmem.c
kmem	test_kMemPartitionAlloc_and_free_manage_free_list_correctly	MEM_RQ_04, MEM_RQ_05	tests/qemu/apps/ut_kmem.c
kmem	test_kMemPartitionAlloc_detects_invalid_object	MEM_RQ_07	tests/qemu/apps/ut_kmem.c
kmem	test_kMemPartitionFree_reuses_most_recently_freed_block_first	MEM_RQ_06	tests/qemu/apps/ut_kmem.c
ksema	test_kSemaphoreInit_sets_expected_fields	SEMA_RQ_01	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreInit_validates_inputs_and_sets_faults	SEMA_RQ_02	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePend_consumes_available_count	SEMA_RQ_03	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePend_respects_no_wait_and_isr_rules	SEMA_RQ_04, SEMA_RQ_05	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePend_can_time_out	SEMA_RQ_06	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePost_wakes_highest_priority_waiter	SEMA_RQ_07, SEMA_RQ_10	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePost_rejects_invalid_objects	SEMA_RQ_16	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePost_binary_saturates_without_waiters	SEMA_RQ_08	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePost_counting_saturates_at_max_without_waiters	SEMA_RQ_09	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePost_waiters_take_precedence_over_counter_increment	SEMA_RQ_07	tests/qemu/apps/ut_ksema.c

Module	Test case	REQ ID(s)	Source file
ksema	test_kSemaphoreFlush_schedules_highest_priority_blocked_task	SEMA_RQ_10	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreFlush_returns_empty_waiting_queue_when_nobody_waits	SEMA_RQ_12	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreFlush_broadcast_releases_all_waiters_and_clears_timeouts	SEMA_RQ_11	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreFlush_from_isr_uses_deferred_path_and_releases_waiters	SEMA_RQ_13	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphorePend_rejects_invalid_timeout	SEMA_RQ_14	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreQuery_reports_waiters_or_count	SEMA_RQ_15	tests/qemu/apps/ut_ksema.c
ksema	test_kSemaphoreQuery_reports_negative_waiter_count_magnitude	SEMA_RQ_15	tests/qemu/apps/ut_ksema.c
kmutex	test_kMutexInit_accepts_valid_protocol_and_initialises_fields	MUTX_RQ_01	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexInit_rejects_invalid_protocol_flags	MUTX_RQ_01	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexLock_applies_transitive_priority_inheritance	MUTX_RQ_07	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexLock_rejects_invalid_timeout_when_contended	MUTX_RQ_06	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexLock_nowait_reports_mutex_locked	MUTX_RQ_02	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexLock_rejects_recursive_lock	MUTX_RQ_03	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexUnlock_rejects_unlock_when_mutex_not_locked	MUTX_RQ_04	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexUnlock_rejects_non_owner	MUTX_RQ_05	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexLock_without_inheritance_does_not_raise_owner_priority	MUTX_RQ_08	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexQuery_reports_lock_state	MUTX_RQ_10	tests/qemu/apps/ut_kmutex.c
kmutex	test_kMutexUnlock_handover_transfers_ownership_and_restores_priority	MUTX_RQ_09	tests/qemu/apps/ut_kmutex.c
kmesgq	test_kMesgQueueInit_validates_supported_message_sizes	MSGQ_RQ_01	tests/qemu/apps/ut_kmesgq.c
kmesgq	test_kMesgQueueSendRecv_is_by_copy_and_preserves_fifo_order	MSGQ_RQ_02, MSGQ_RQ_03	tests/qemu/apps/ut_kmesgq.c
kmesgq	test_kMesgQueue_no_wait_returns_empty_and_full_without_blocking	MSGQ_RQ_04, MSGQ_RQ_05	tests/qemu/apps/ut_kmesgq.c
kmesgq	test_kMesgQueue_owner_only_receiver_is_enforced	MSGQ_RQ_06	tests/qemu/apps/ut_kmesgq.c

Module	Test case	REQ ID(s)	Source file
kmesgq	test_kMesgQueue_notify_callback_runs_on_successful_send	MSGQ_RQ_07	tests/qemu/apps/ut_kmesgq.c
kmesgq	test_kMesgQueueReset_clears_queue_and_wakes_waiter	MSGQ_RQ_08	tests/qemu/apps/ut_kmesgq.c
kmesgq	test_kMesgQueueReset_in_isr_defers_when_waiters_exist	MSGQ_RQ_09	tests/qemu/apps/ut_kmesgq.c
kport	test_kPortInit_requires_metadata_words_and_sets_server_endpoint	PORT_RQ_01	tests/qemu/apps/ut_kport.c
kport	test_kPortOwner_is_single_assignment	PORT_RQ_02	tests/qemu/apps/ut_kport.c
kport	test_kPortSend_writes_sender_metadata_and_null_reply_route	PORT_RQ_03	tests/qemu/apps/ut_kport.c
kport	test_kPortRecv_rejects_non_owner_receiver	PORT_RQ_04	tests/qemu/apps/ut_kport.c
kport	test_kPortRecv_and_server_done_move_ready_server_between_priority_queues	PORT_RQ_05, PORT_RQ_06	tests/qemu/apps/ut_kport.c
kport	test_kPortRecv_priority_adoption_prevents_unrelated_task_inversion	PORT_RQ_07	tests/qemu/apps/ut_kport.c
kport	test_kPortSendRecv_priority_adoption_prevents_unrelated_task_inversion	PORT_RQ_05, PORT_RQ_06, PORT_RQ_07, PORT_RQ_09, PORT_RQ_13	tests/qemu/apps/ut_kport.c
kport	test_kPort_multiple_clients_priority_range_preserves_invariants	PORT_RQ_08	tests/qemu/apps/ut_kport.c
kport	test_kPortSendRecv_initializes_reply_box_and_sets_call_metadata	PORT_RQ_09	tests/qemu/apps/ut_kport.c
kport	test_kPortSendRecv_rejects_reply_box_owned_by_other_task	PORT_RQ_10	tests/qemu/apps/ut_kport.c
kport	test_kPortReply_requires_reply_box_in_message_metadata	PORT_RQ_11	tests/qemu/apps/ut_kport.c
kport	test_kPortReplyDone_demotes_server_even_when_reply_fails	PORT_RQ_12	tests/qemu/apps/ut_kport.c
kport	test_kPortSend_and_recv_adopts_sender_priority	PORT_RQ_05, PORT_RQ_06	tests/qemu/apps/ut_kport.c
kport	test_kPortReplyDone_posts_reply_and_demotes_server	PORT_RQ_13	tests/qemu/apps/ut_kport.c
kport	test_kPortSendRecv_multiple_clients_sum_reply	PORT_RQ_14	tests/qemu/apps/ut_kport.c



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